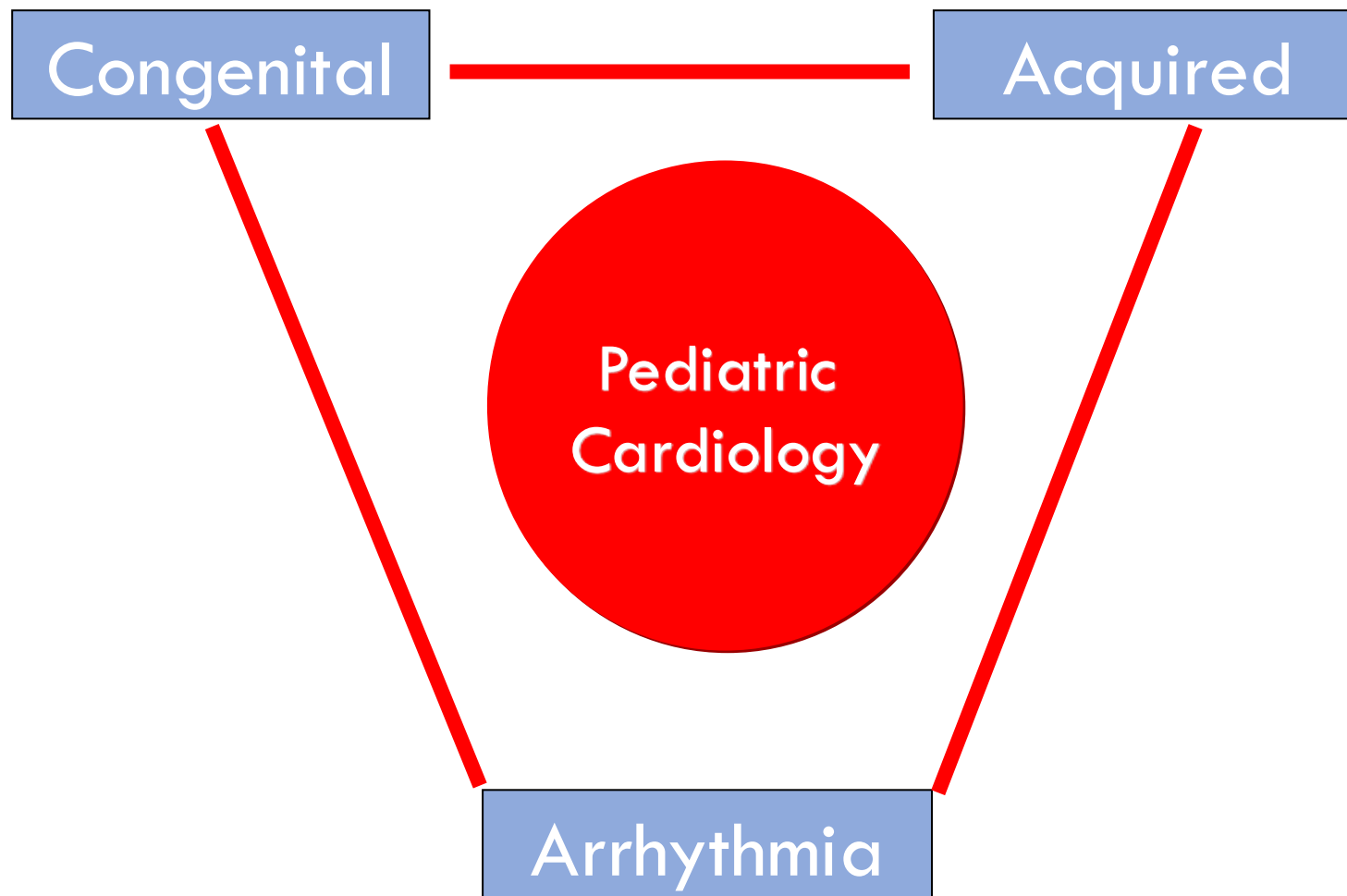




Acyanotic Heart disease In pediatric





Agenda



CHD: VSD, PDA, ASD
Congestive heart failure(CHF)



Clinical presentation



Pathophysiology



Basic investigation



Etiology



Treatment



Objective

1. บอกแนวทางในการประเมิน ผู้ป่วยเด็กที่มีปัญหาเรื่องหัวใจพิการแต่กำเนิดชนิดไม่เขียวและภาวะหัวใจวายได้ถูกต้อง
2. บอกพยาธิกำเนิด พยาธิสรีรวิทยา ลักษณะทางคลินิกของผู้ป่วยเด็กที่มีปัญหาเรื่องหัวใจพิการแต่กำเนิดชนิดไม่เขียวและหัวใจวายได้ถูกต้อง



ข้อใดไม่ใช่โรคหัวใจพิการแต่กำเนิดชนิดไม่เขียว (acyanotic heart disease)

1. Ventricular septal defect: **VSD** ผันหัวใจห้องล่างรั่ว
2. Patent ductus arteriosus: **PDA** เส้นเลือดเกิน
3. Atrial septal defect: **ASD** ผันหัวใจห้องบนรั่ว
4. Myocarditis: ดกกล้ามเนื้อหัวใจอักเสบ

โรคหัวใจชนิดใดพบมากที่สุด

1. Patent ductus arterioles: **PDA** เส้นเลือดเกิน
2. ~~Ventricular septal defect: **VSD**~~ dผนังหัวใจห้องล่างรั่ว
3. Atrial septal defect: **ASD** dผนังหัวใจห้องบนรั่ว
4. I don't know

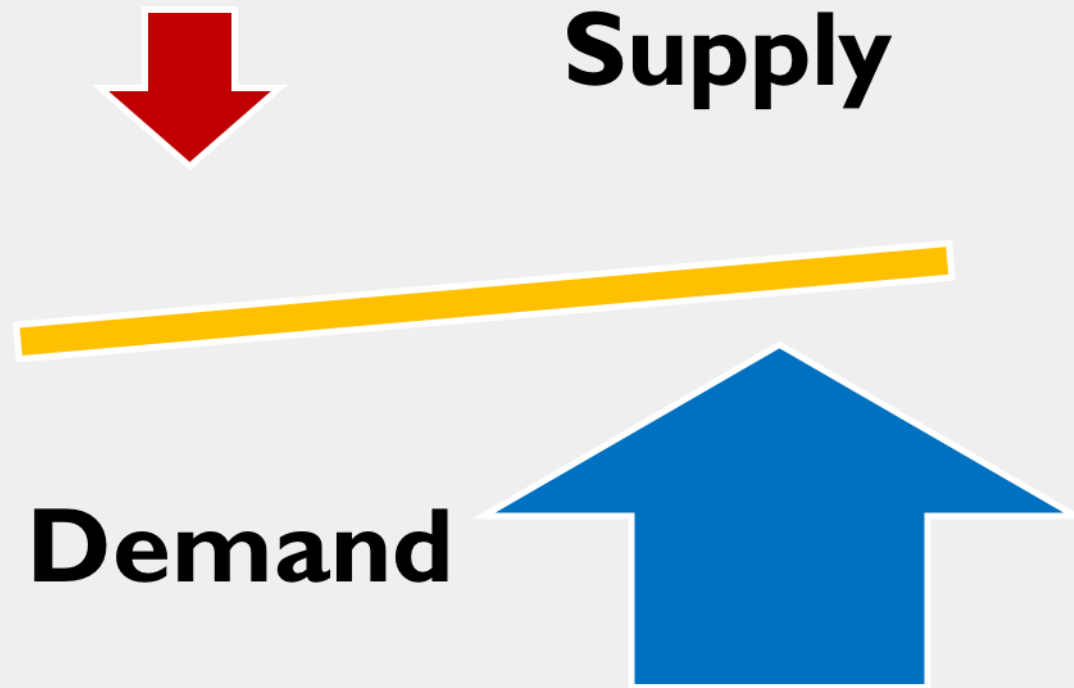
“Question ”

How do you diagnose HF ???

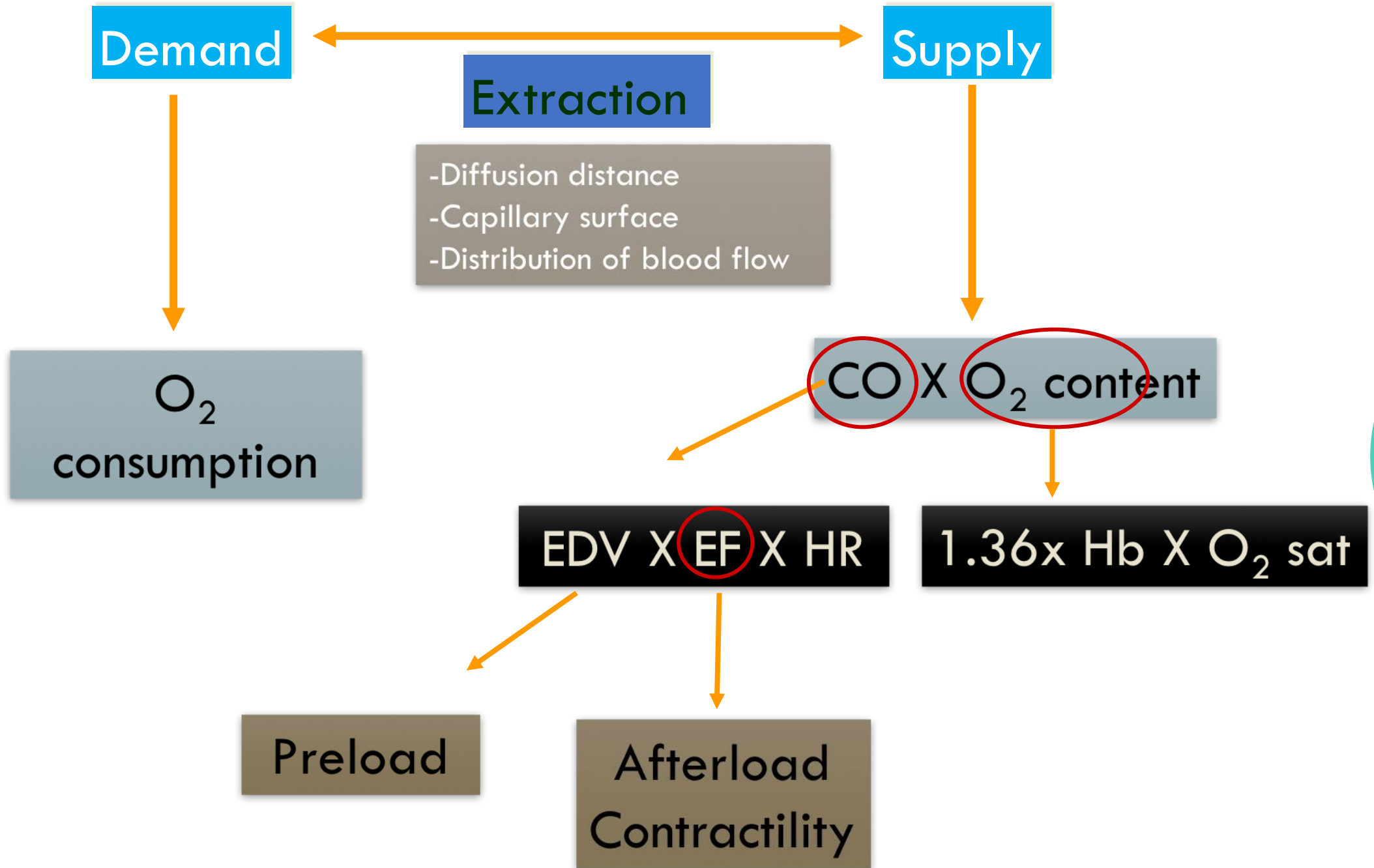
What is cause of HF ???

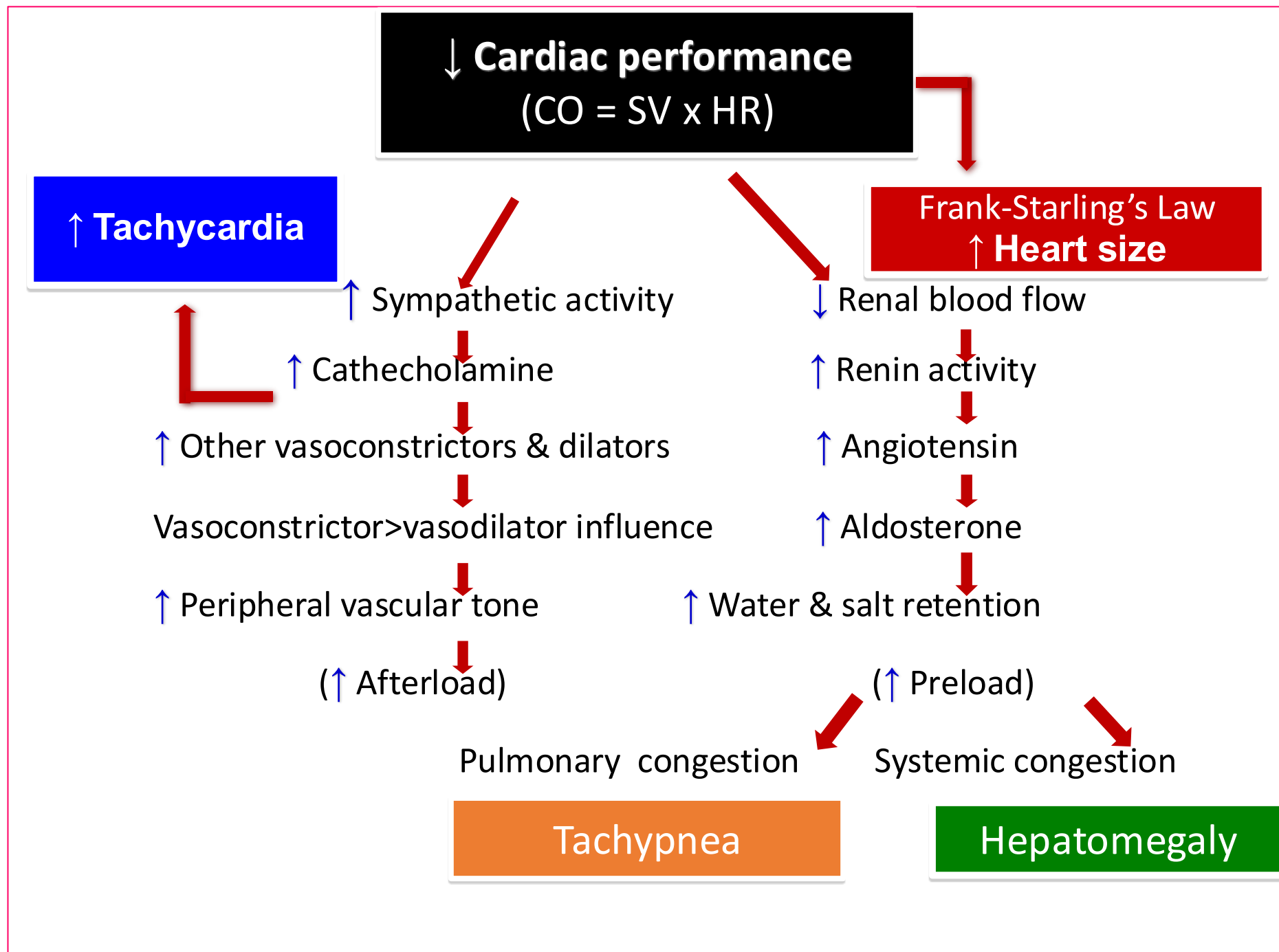
What are the clinical manifestation of HF ???

“Heart Failure is a clinical syndrome”



Definition







Clinical presentation

Infant



Growth failure
Persistent tachycardia
Respiratory distress

Toddlers



Respiratory distress
Poor appetite
Decrease activity

school age



Fatigue
Orthopnea
Poor appetite
Exercise intolerance

adolescents



Dyspnea
Chest pain
Orthopnea
Nausea/vomiting



presentation

อาการใดบ่งบอกภาวะหัวใจล้มเหลวเบื้องต้นในเด็กทารก

1. Apnea: หยุดหายใจ

2. Cough: ไอ

3. Edema: บวม

4. Seizure: ชัก

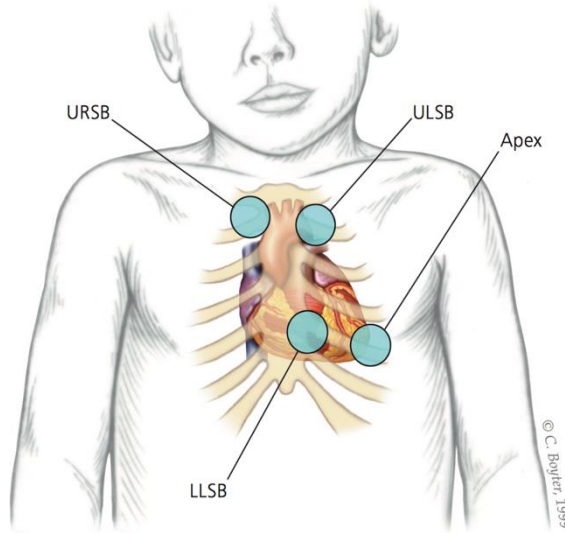
5. Dyspnea on exertion: หายใจลำบาก เหนื่อยง่าย เหงื่อแตกหน้าผาก

Red flag history of CHF



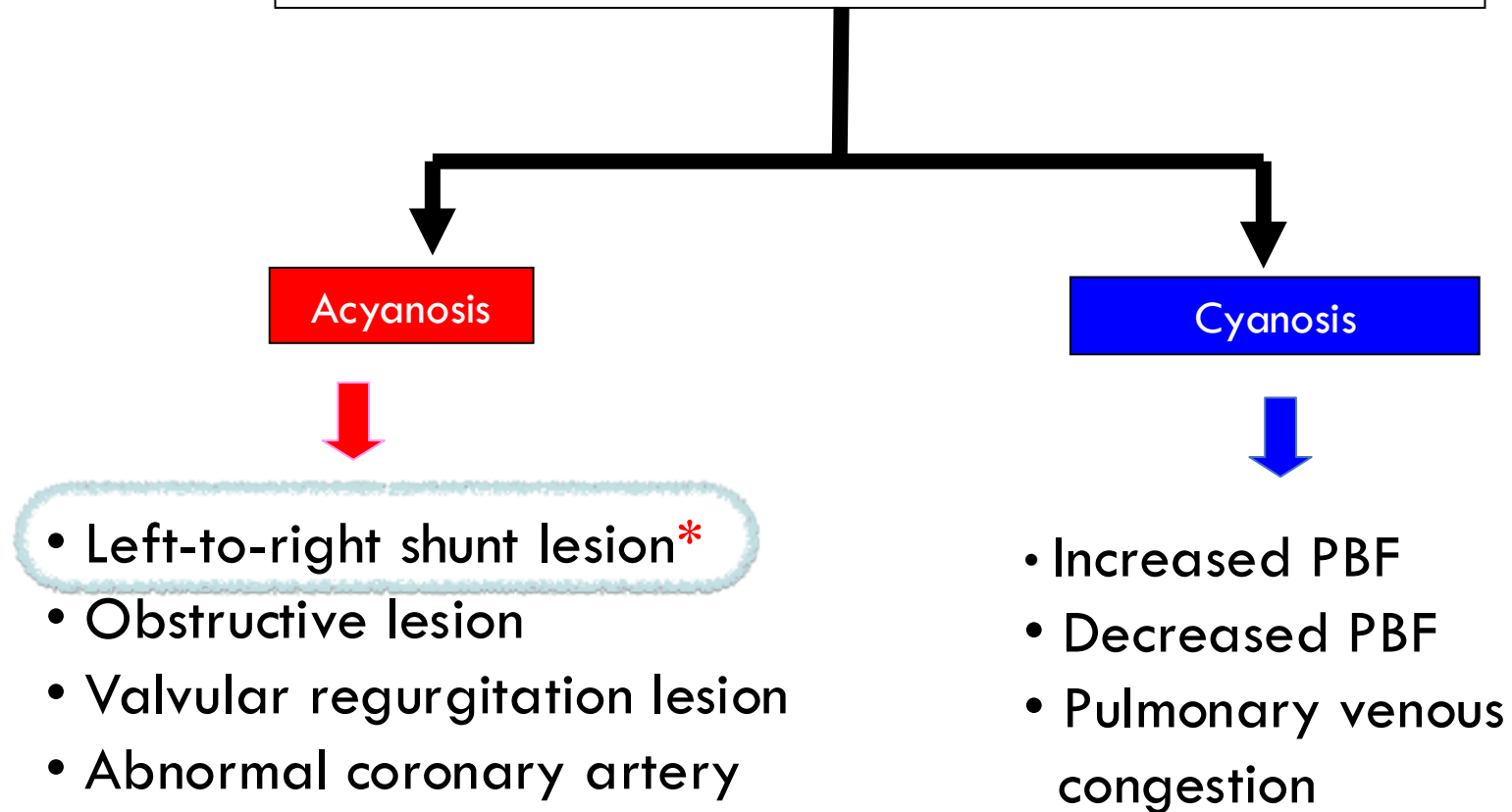
- Poor feeding
- Poor weight gain
- Frequent lower respiratory infection: pneumonia
- Decrease exercise tolerance

Physical examination

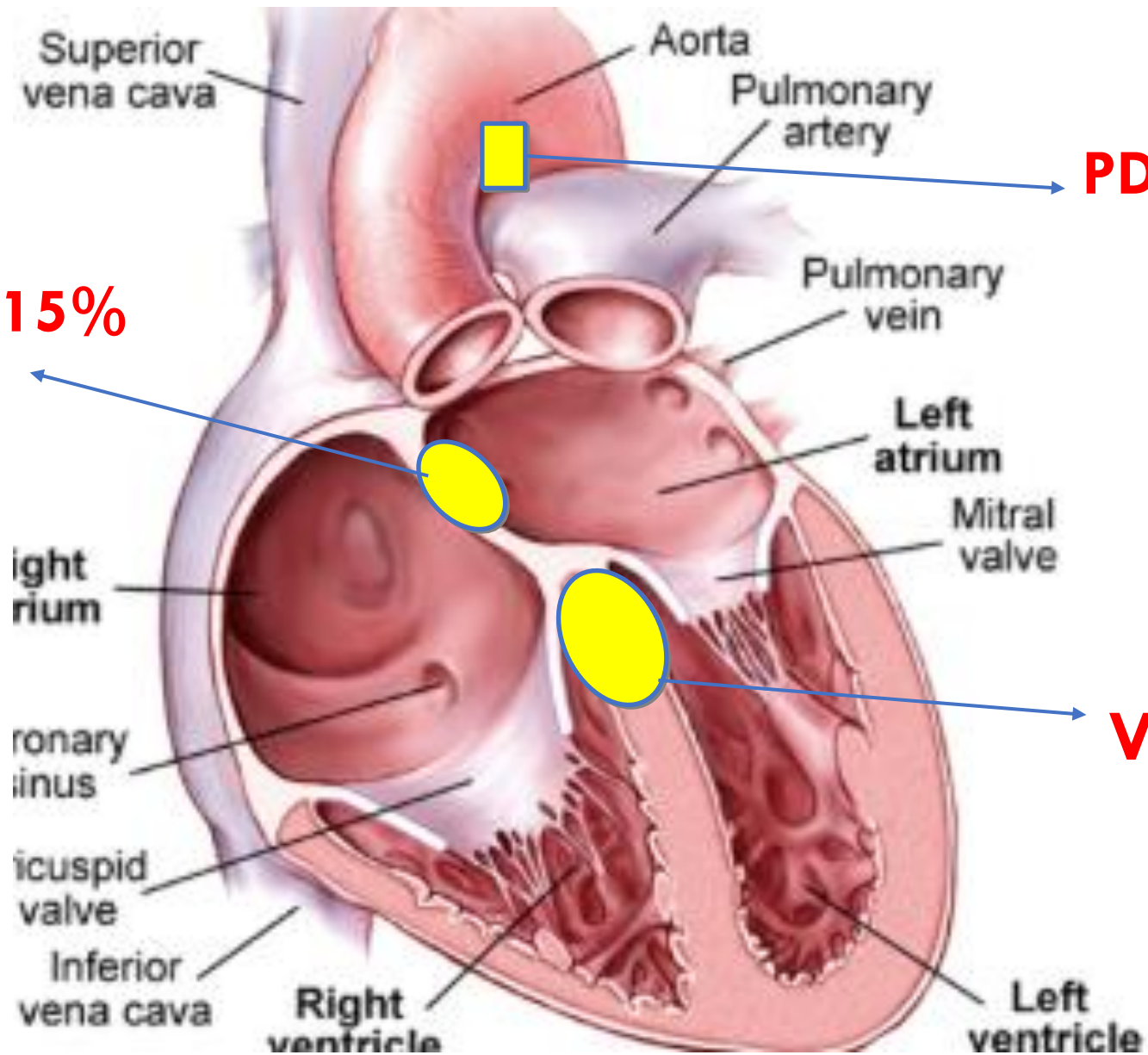


- Syndrome
- Nutritional status: weight, height
- General appearance: well being, respiratory distress
- Color: conjunctiva, oral mucosa
- Chest wall: deformity, bulging, active precordium
- Extremities: deformity, clubbing
- Skin rashes

Congenital Heart Disease (incidence: 1:100 NB)



ASD: 5-15%



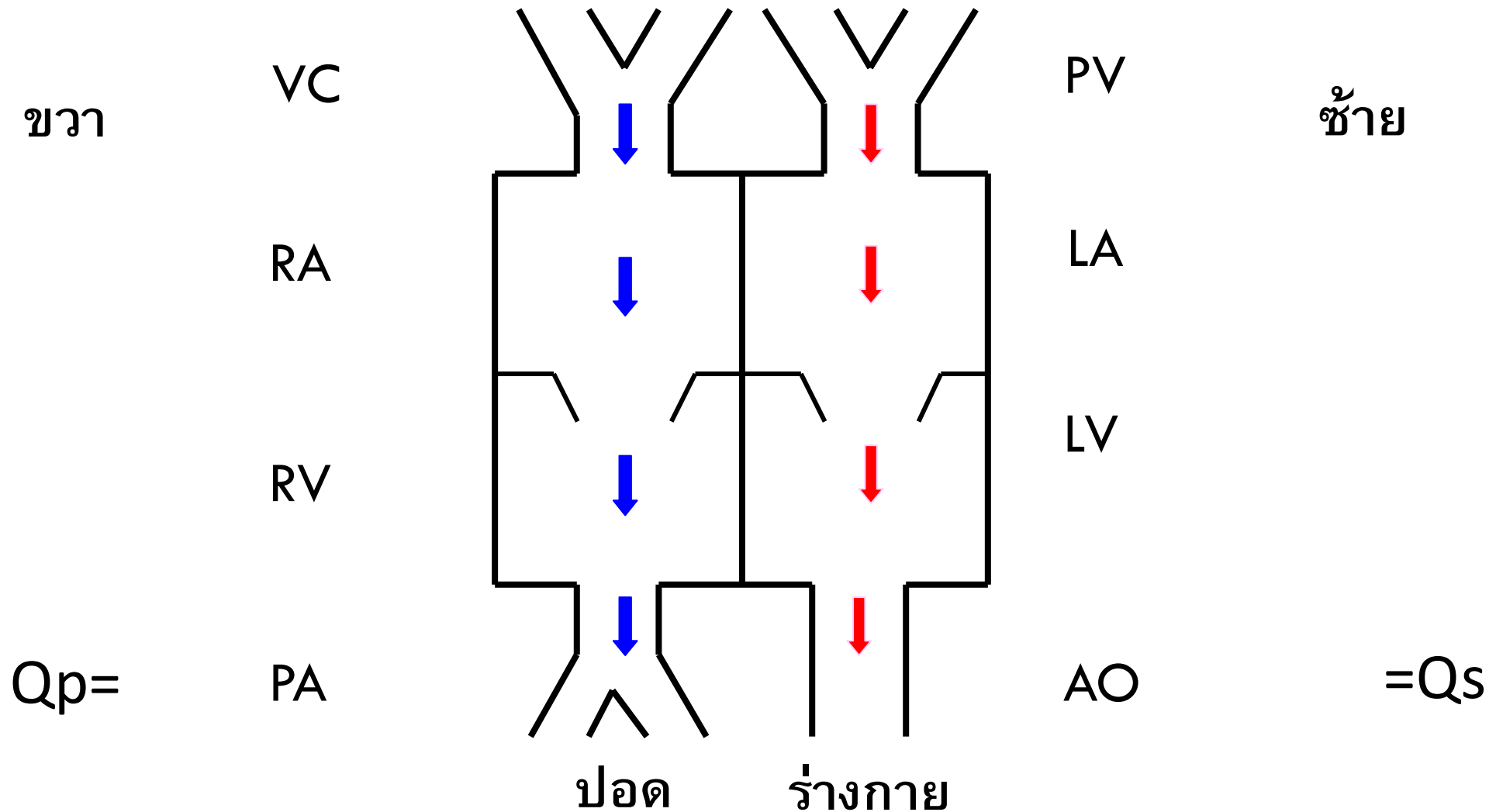
PDA: 5-10%

VSD: 15-30%

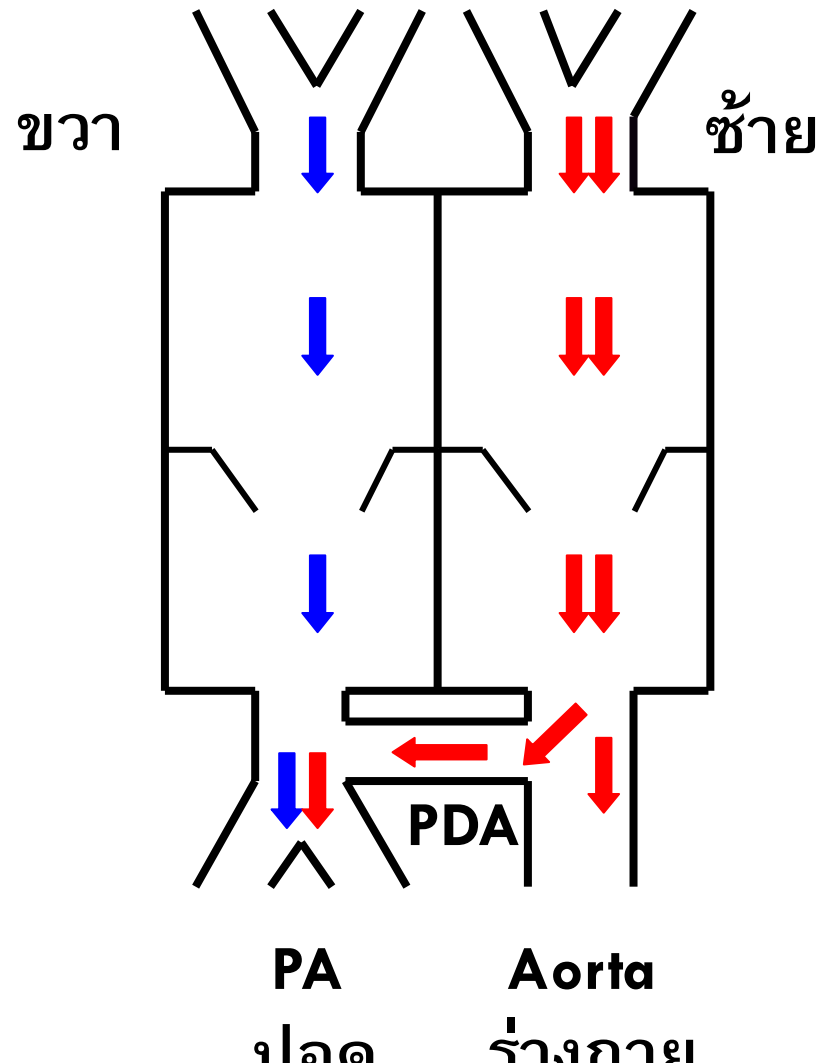


Epidermiology

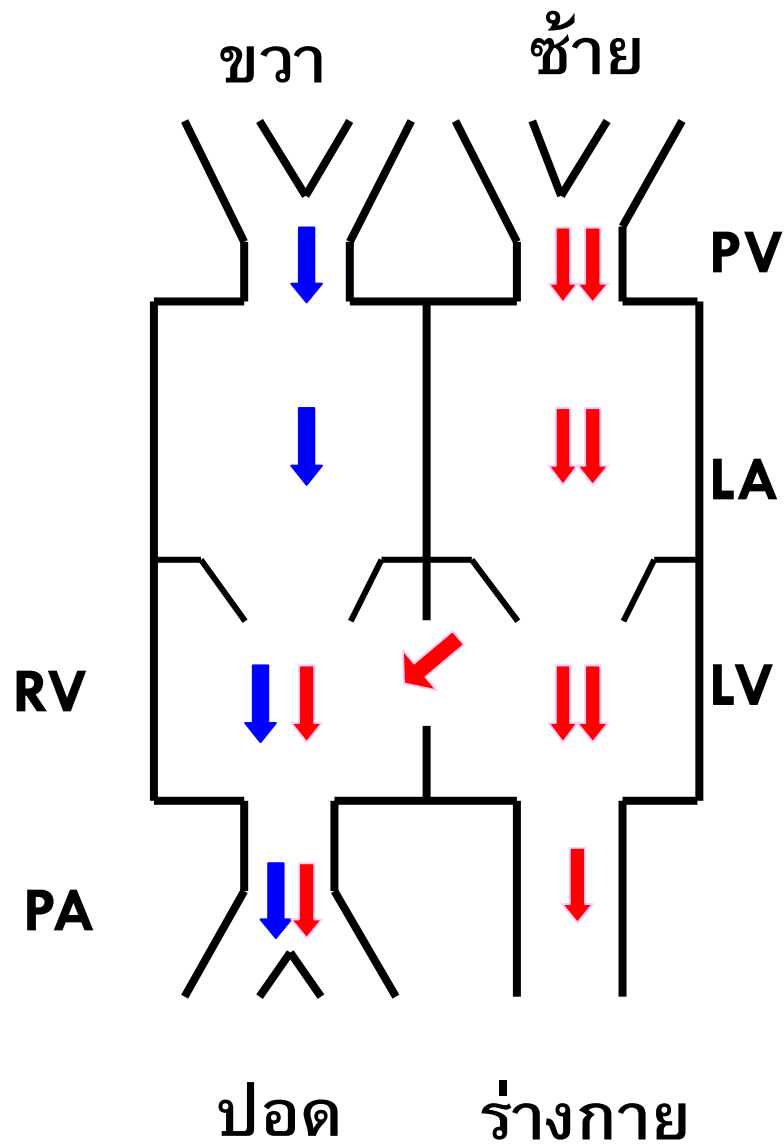
Normal Physiology



Patent ductus arterioles



Ventricular septal defect

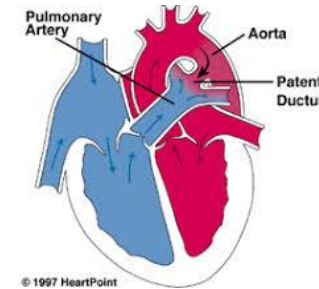
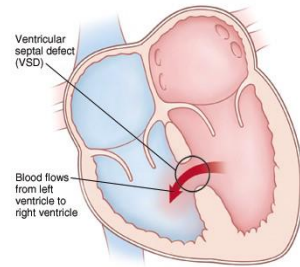


“Volume overload”

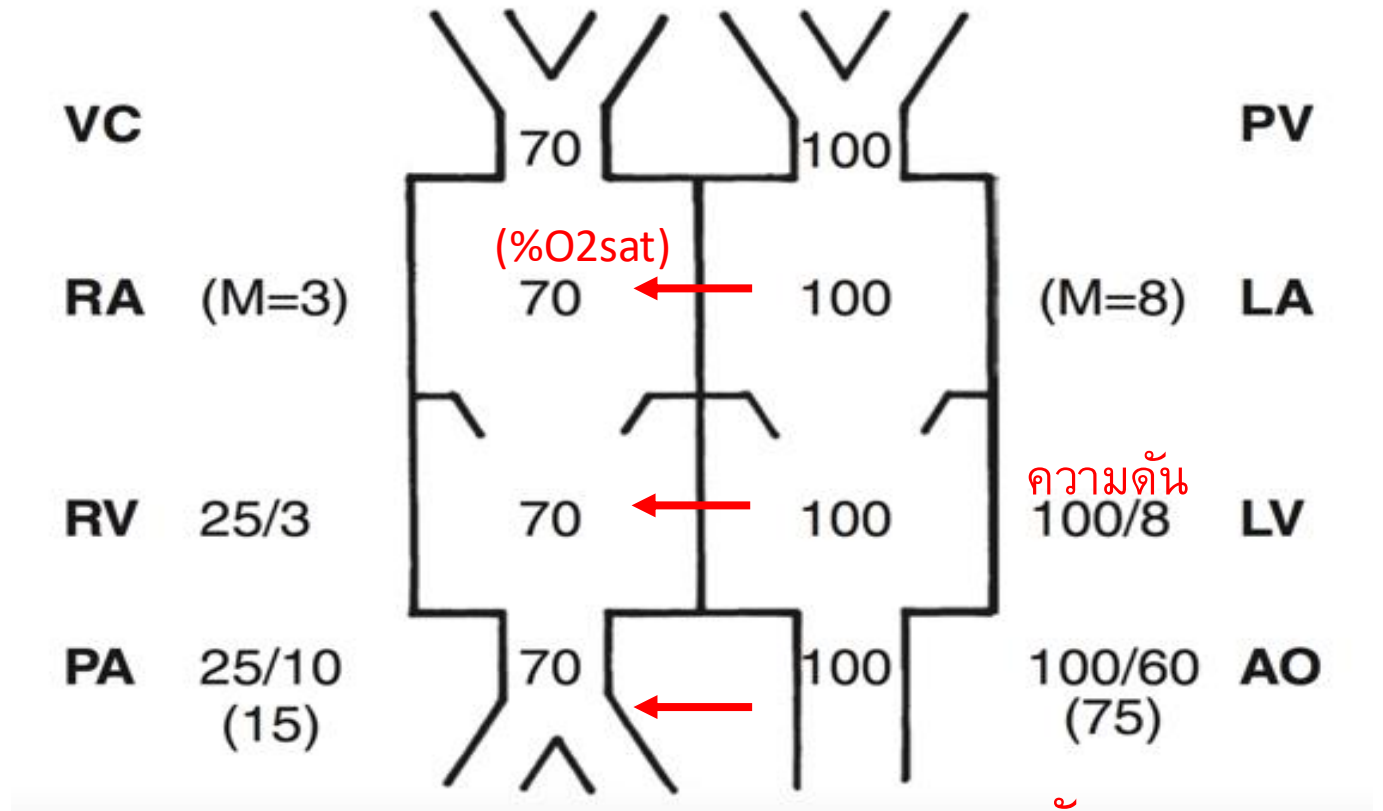
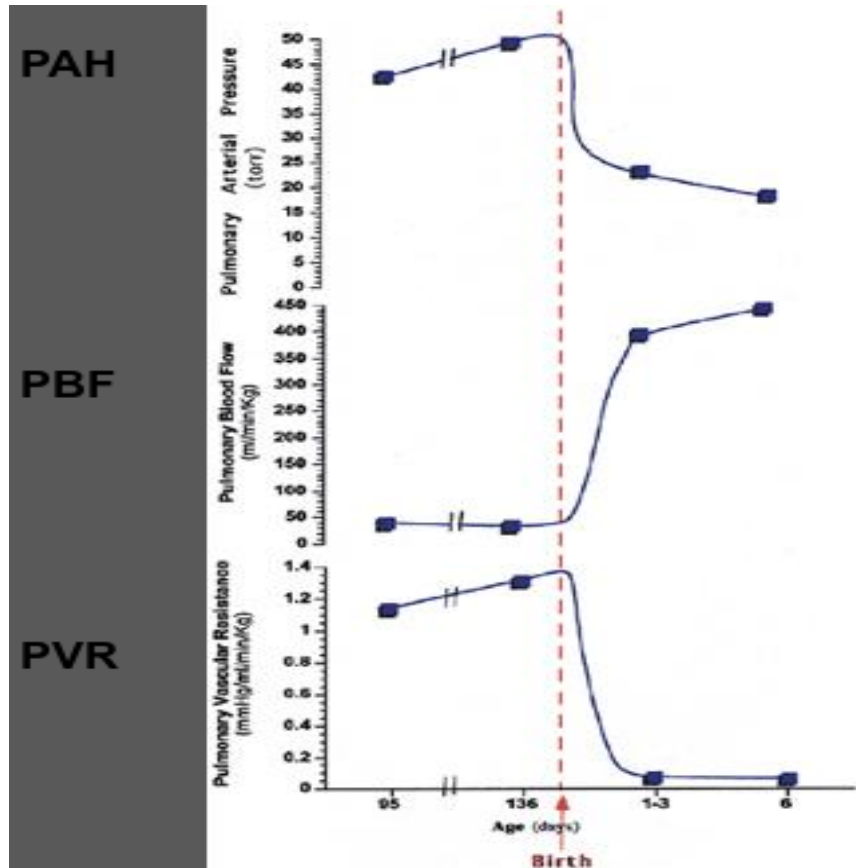
Left to right shunt



VSD ผนังหัวใจห้องล่างรั่ว



PDA เส้นเลือดเกิน



ปอด ร่างกาย

หลังคลอด

What is important factor of HF???

PVR ↓

Size

Ventricular
compliance

Poor
gain
weight

M

Asymptomatic
Heart Murmur

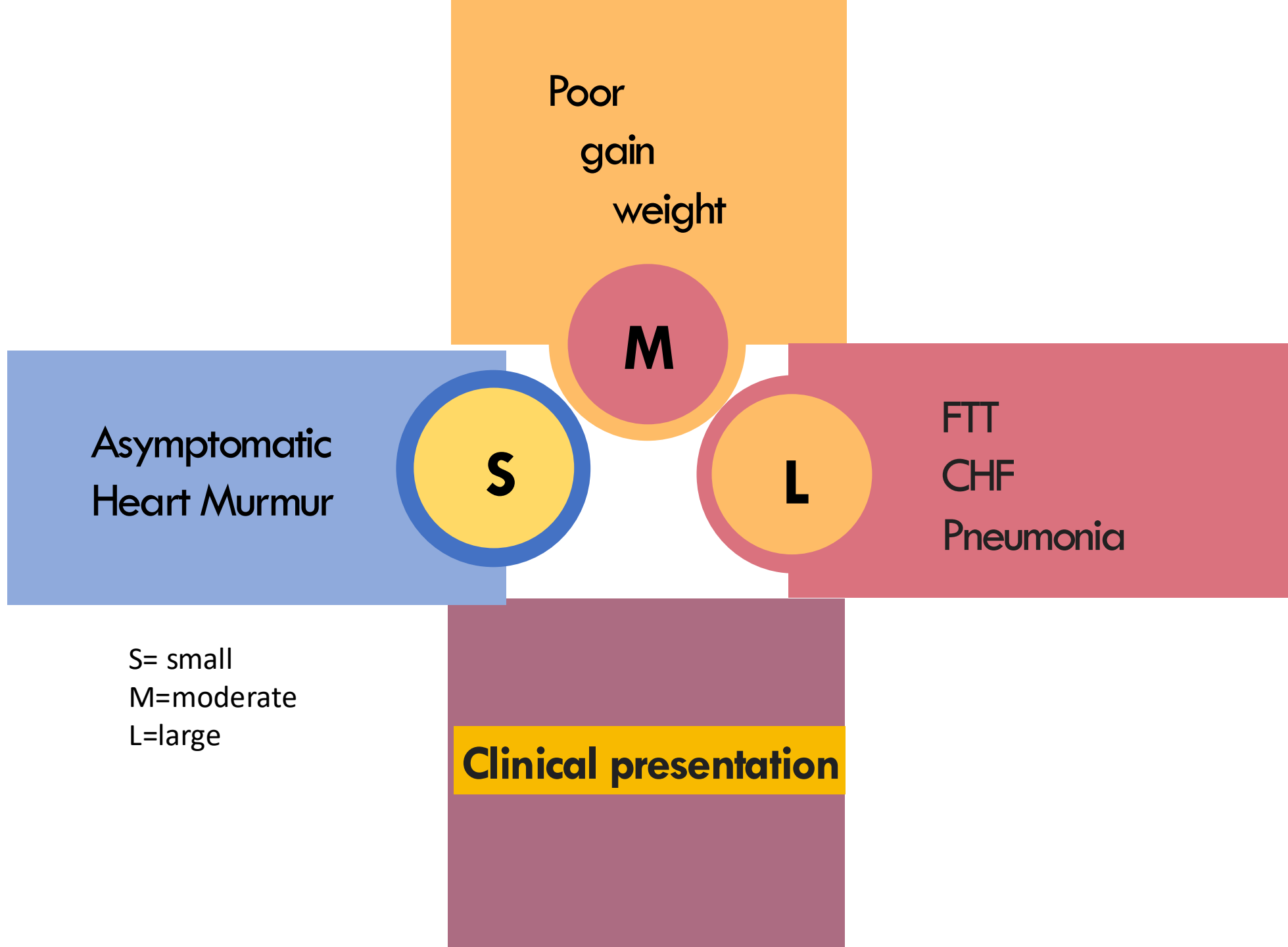
S

L

FTT
CHF
Pneumonia

S= small
M=moderate
L=large

Clinical presentation



Case 1: 2-month-old boy



Dyspnea for 1 wk

V/S: BP 70/45 mmHg, PR 165/min, RR 65/min, O₂ sat 99%

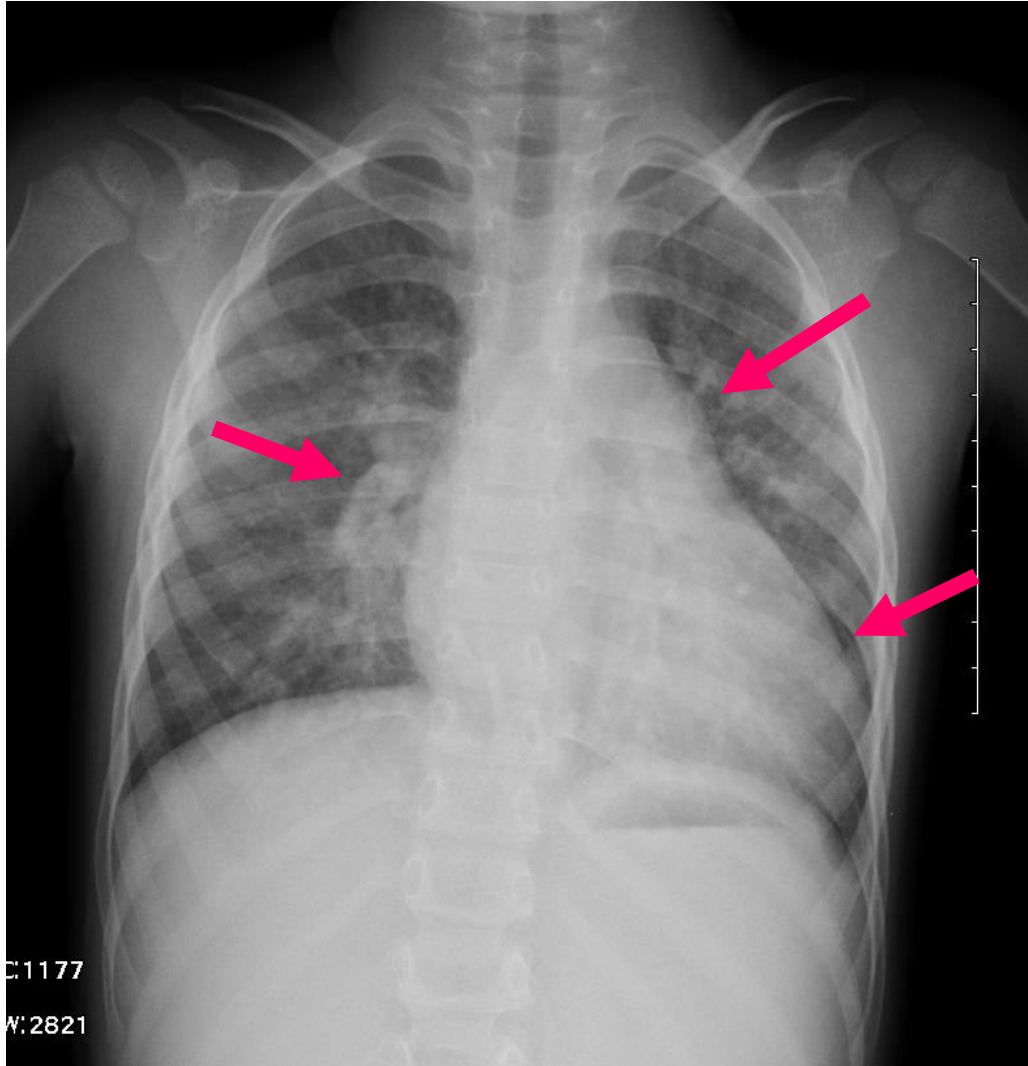
Heart: active precordium,

Lungs: normal equal breath sound

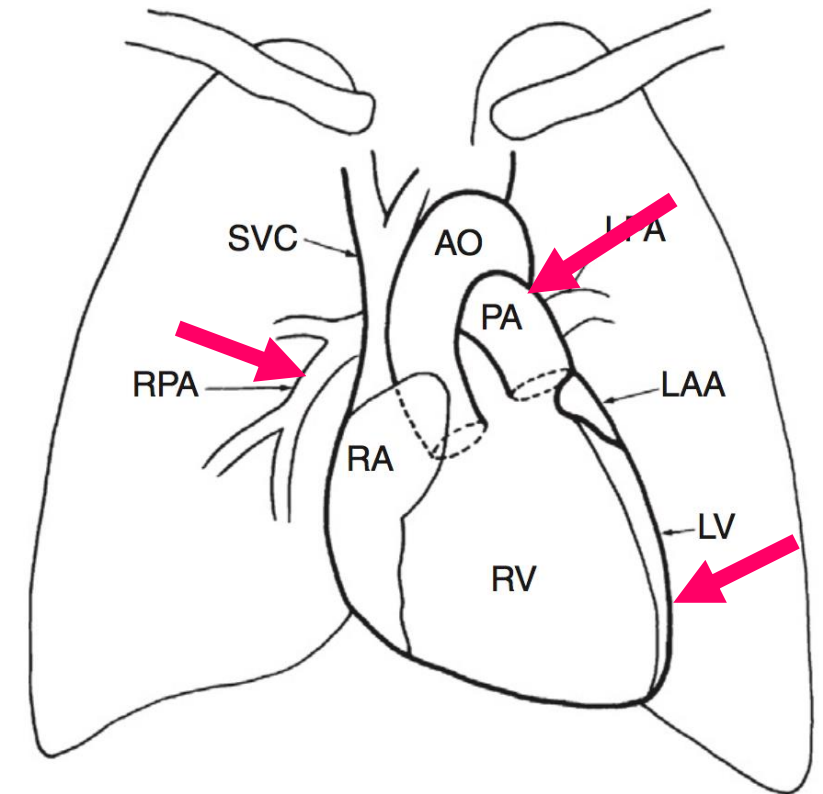
Liver 2 cm BRCM



Cardiomegaly, enlarged MPA, LV enlargement, increased PVM

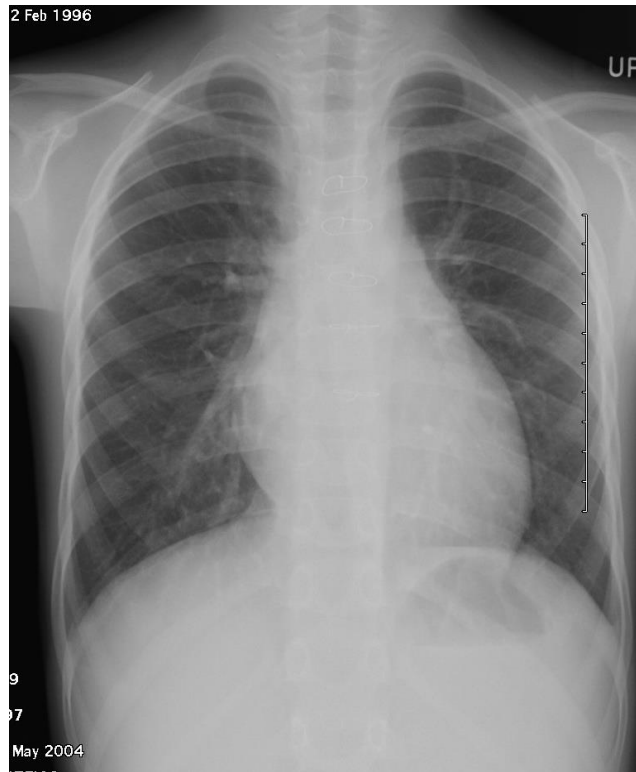


Chest X-ray



Chest X ray

Small VSD



Moderate VSD



Large VSD

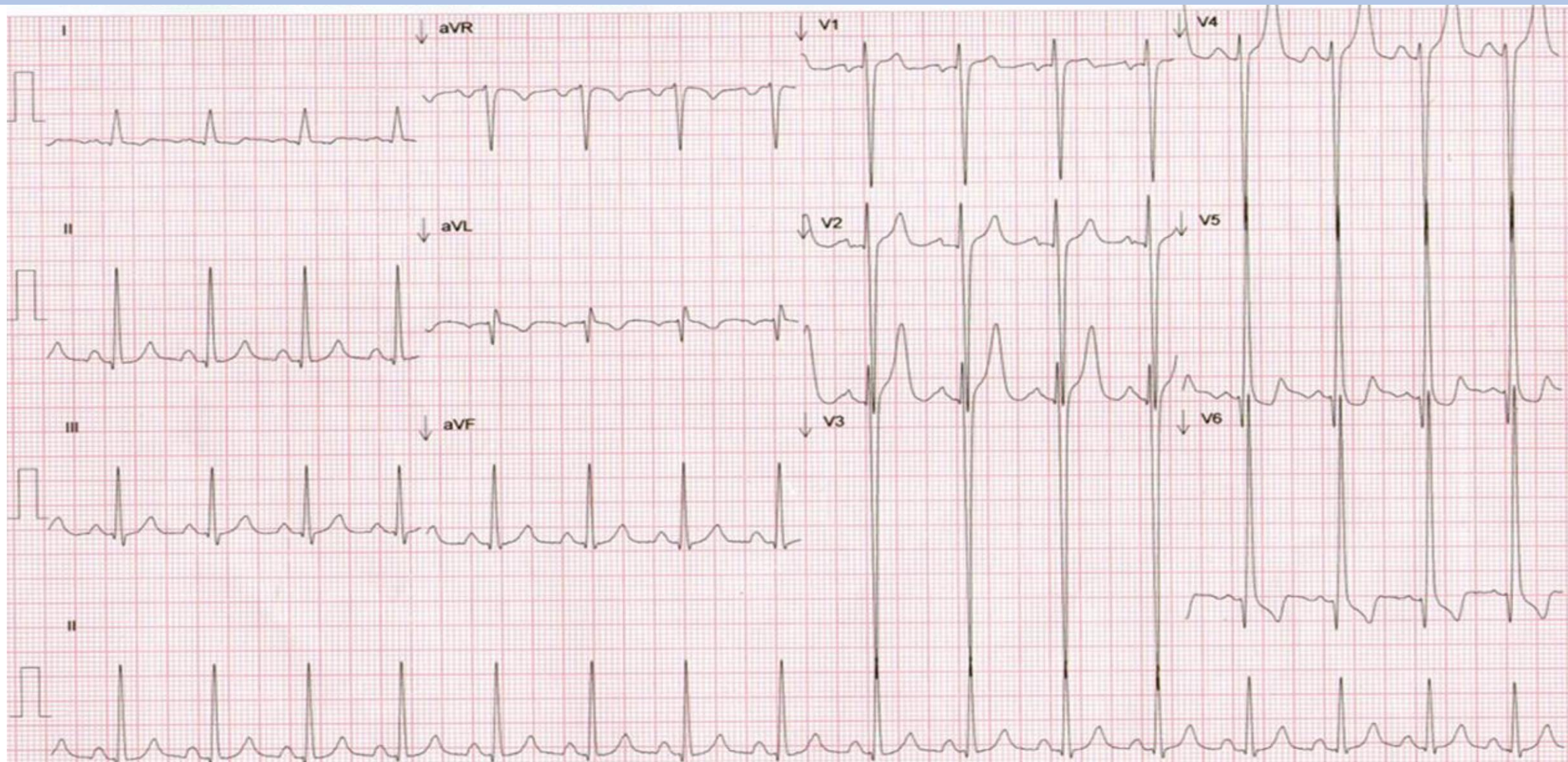


ขนาดเล็ก

ขนาดกลาง

ขนาดใหญ่

EKG

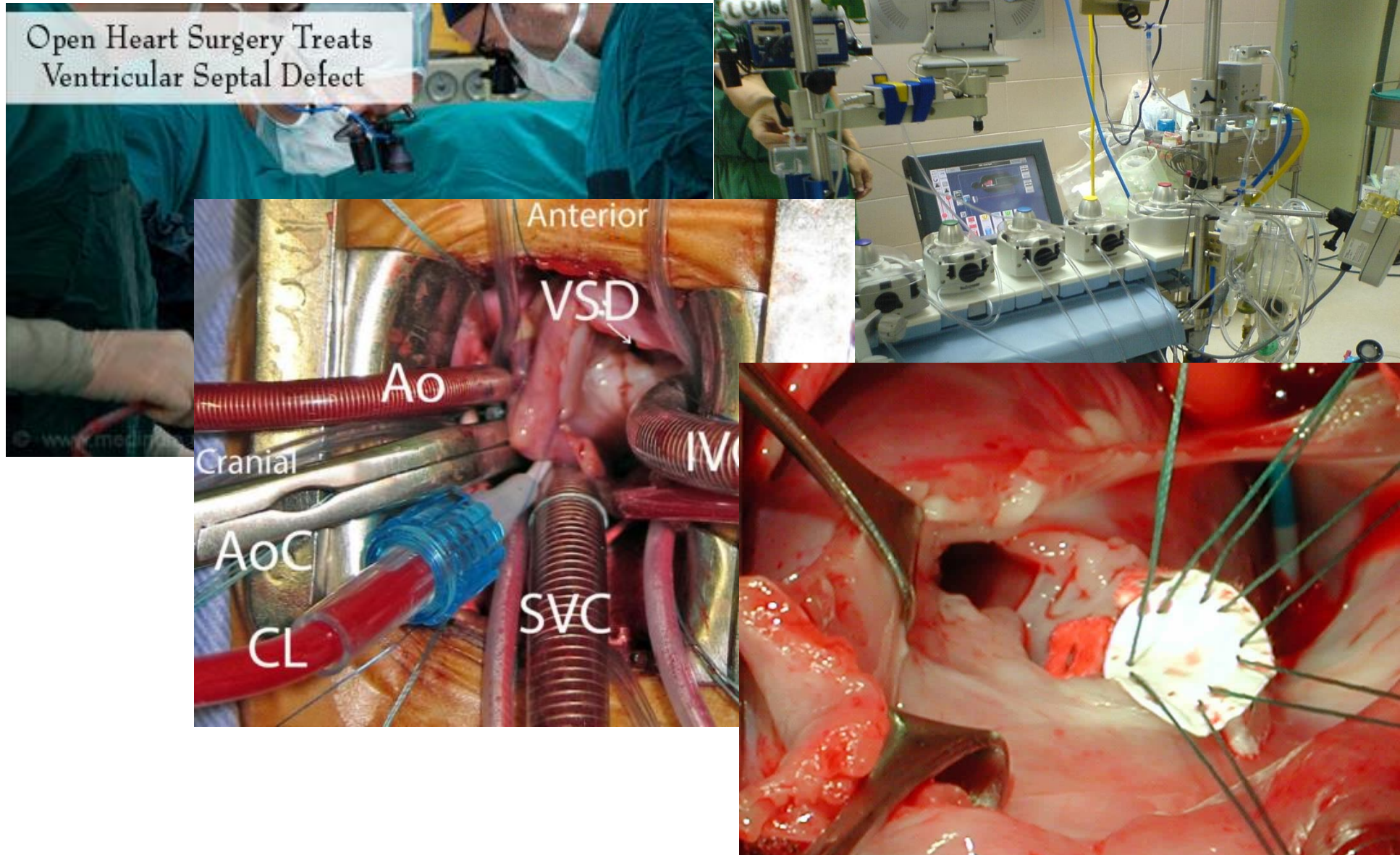


Treatment

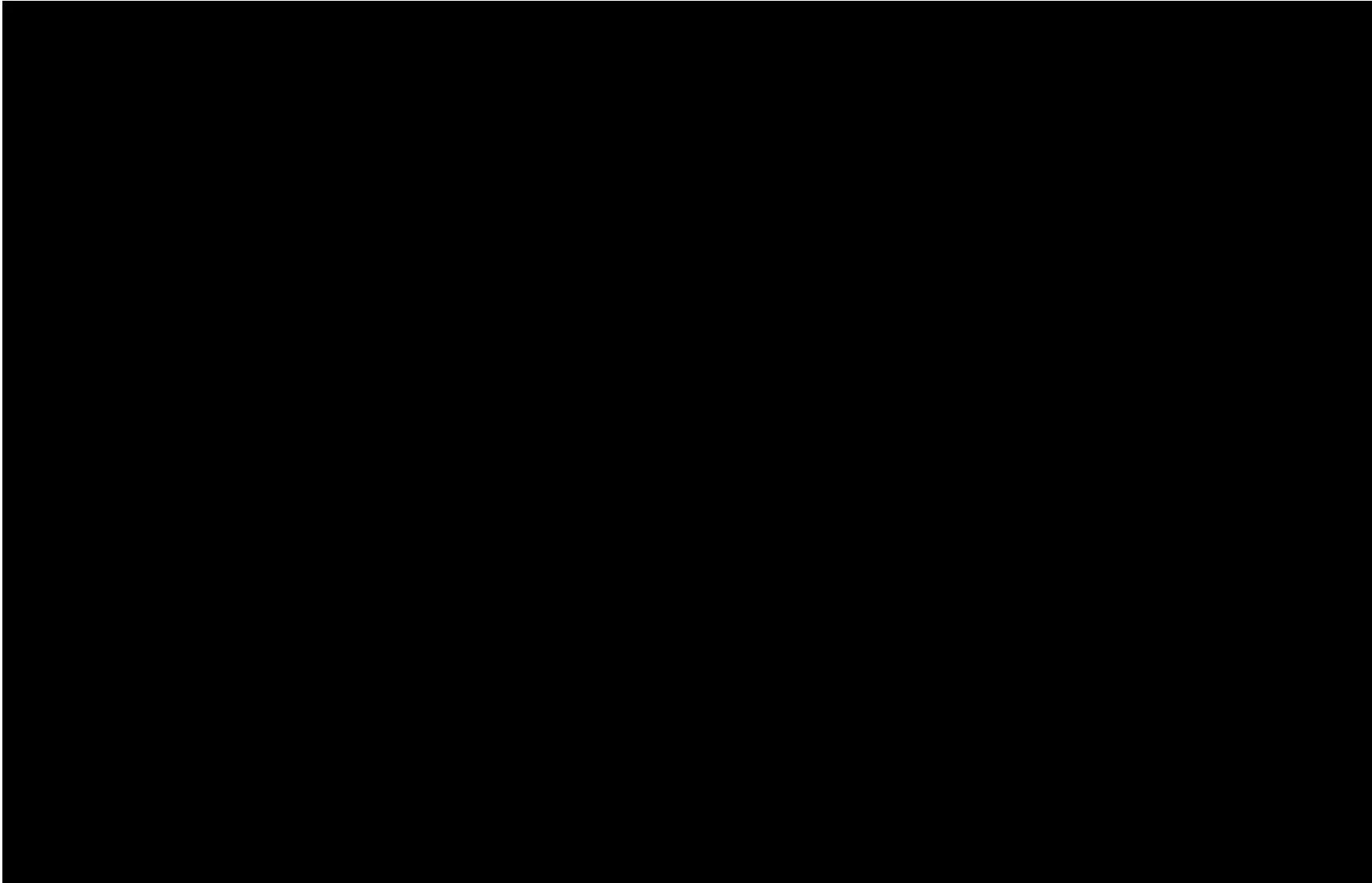
- **Non-surgical Tx**
 - **No symptom:** no medication, dental care
 - **Symptom:** Anti-CHF drug, nutrition
- **VSD closure: Surgical/ device closure**

Open heart surgery: VSD surgical closure

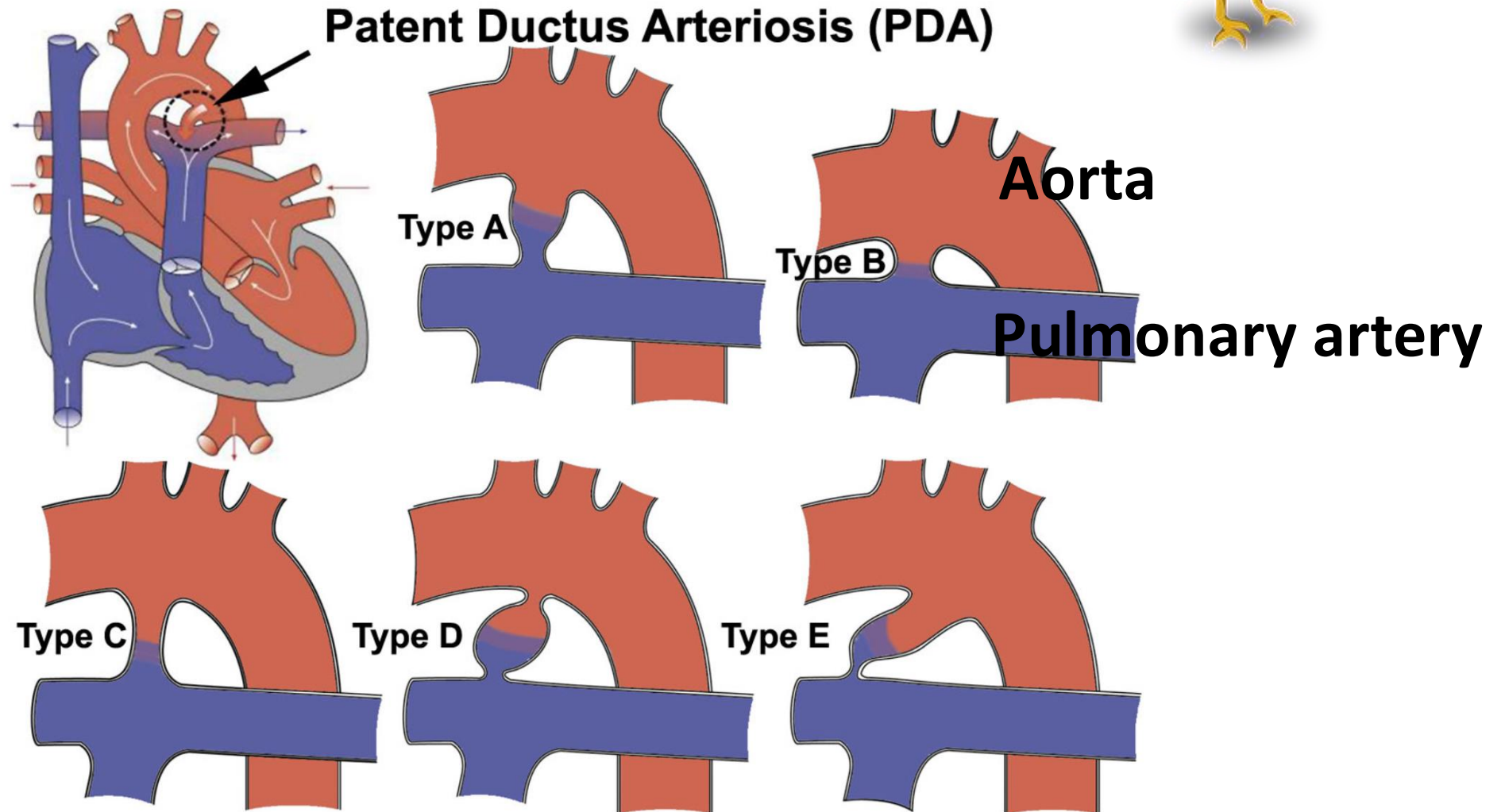
Cardiopulmonary Bypass



Intervention with VSD device

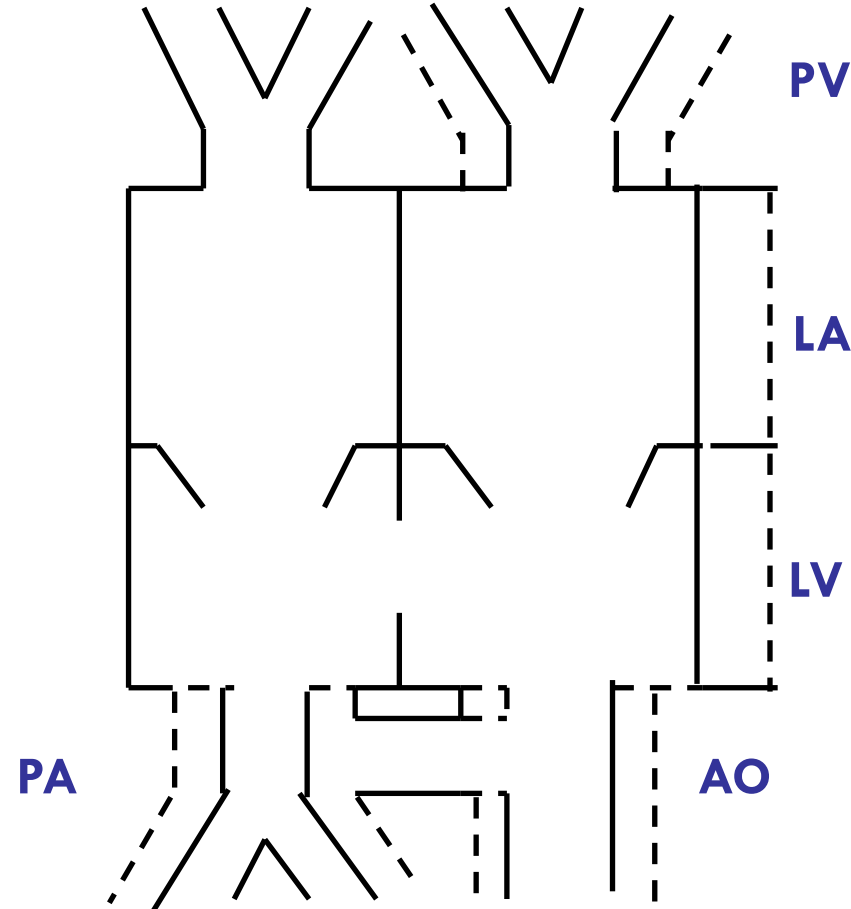
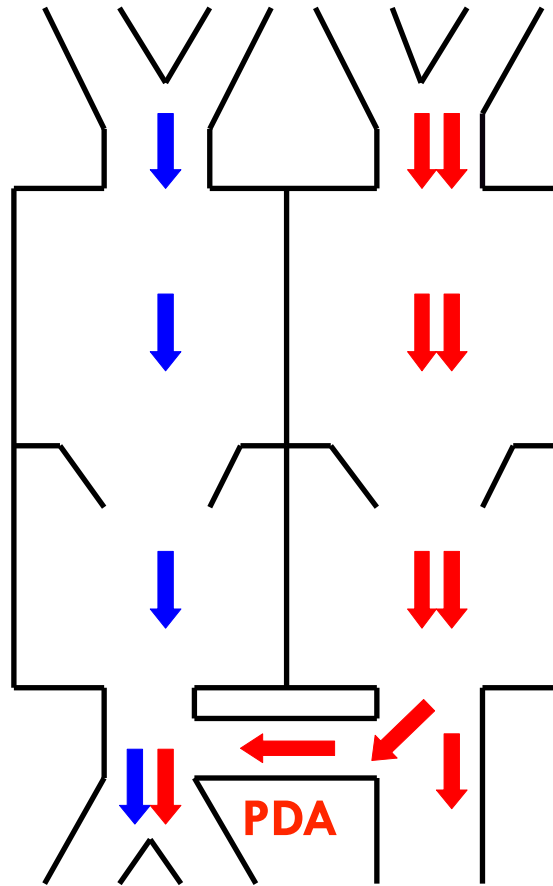


Patent Ductus Arteriosus



Krichenko angiographic classification (1989) PMID2929450

Patent Ductus Arteriosus



$Q_p =$

ปอด

ร่างกาย

$= Q_s$

Poor
gain
weight

M

Asymptomatic
Heart Murmur

S

FTT
CHF
Pneumonia

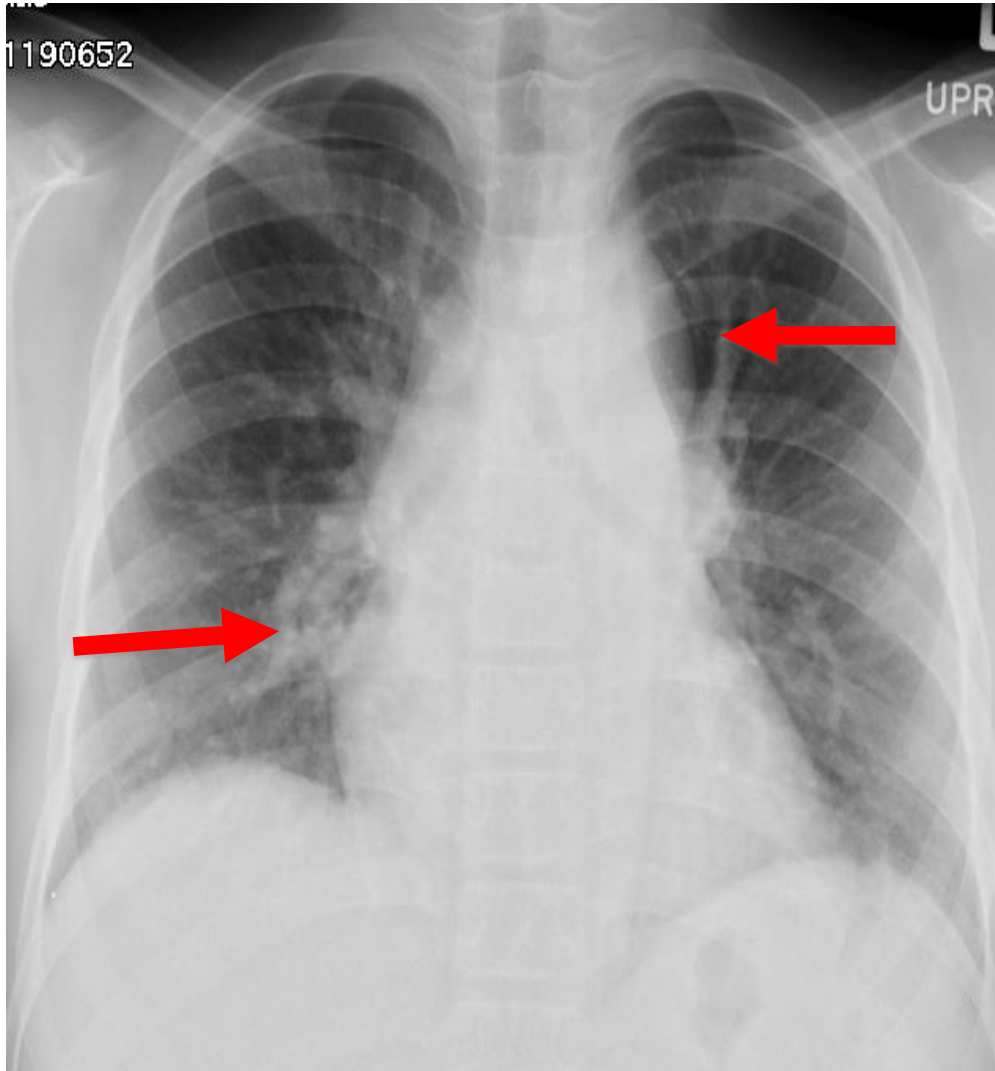
L

Clinical presentation

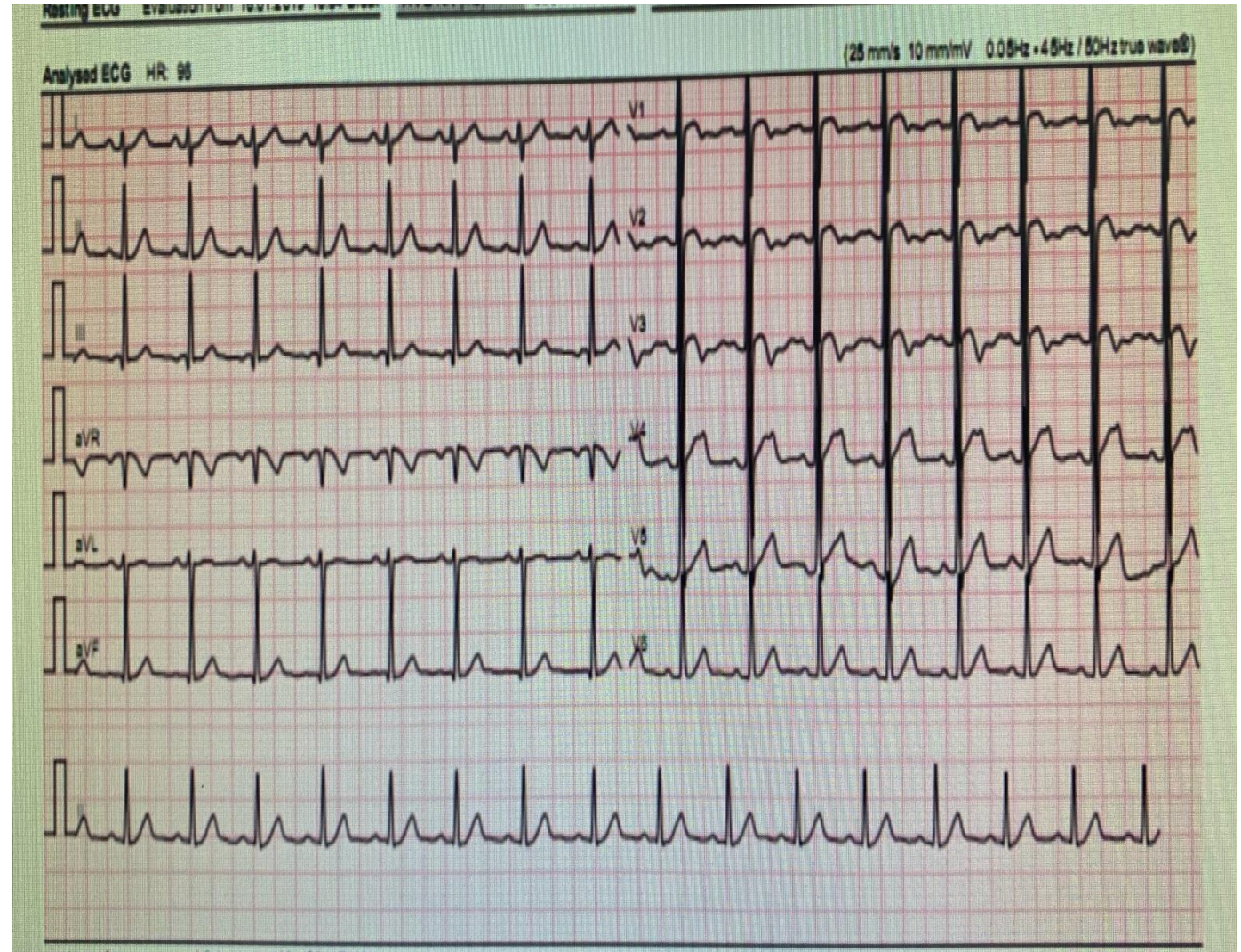
Patent Ductus Arteriosus



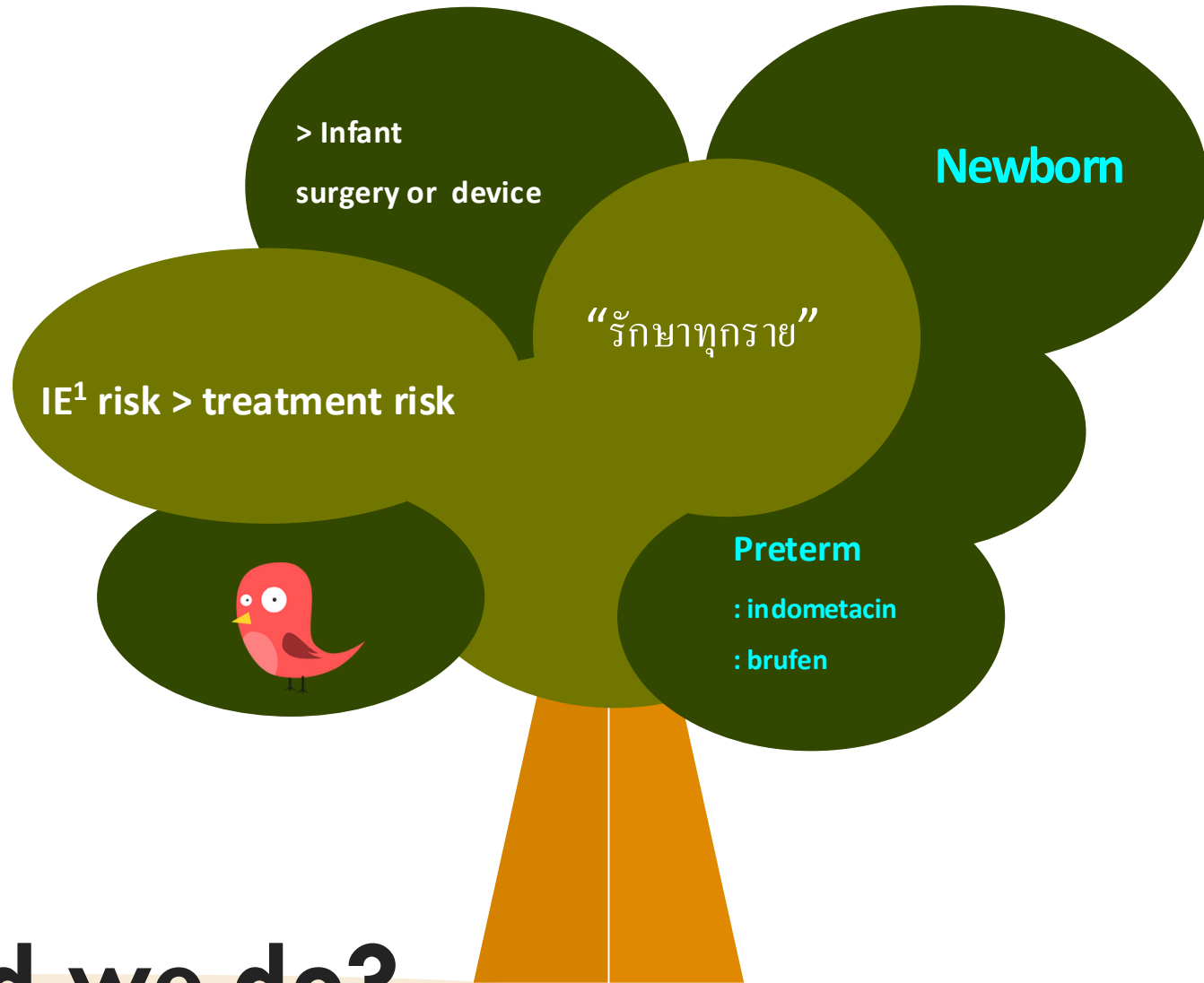
CXR



EKG

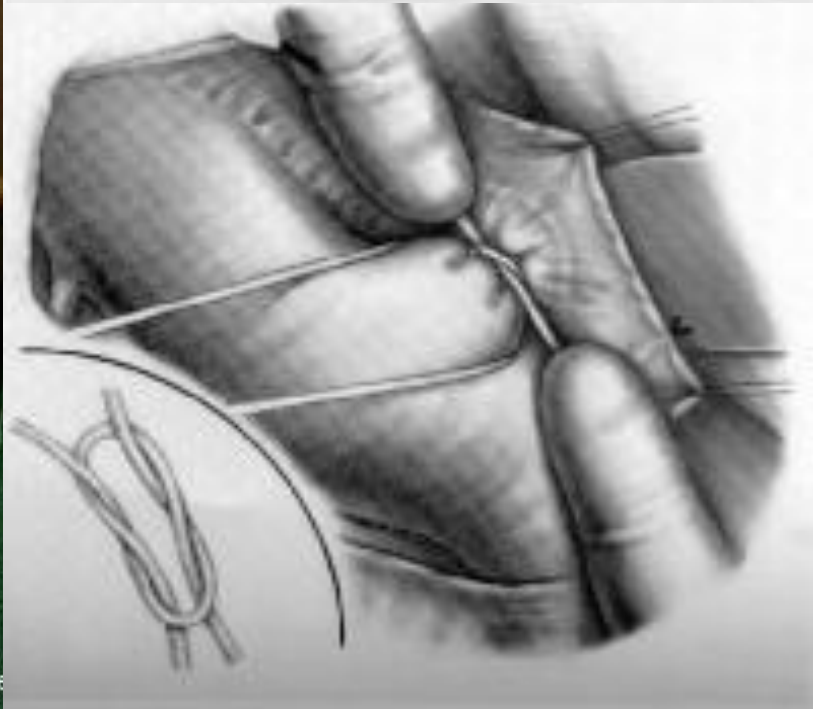
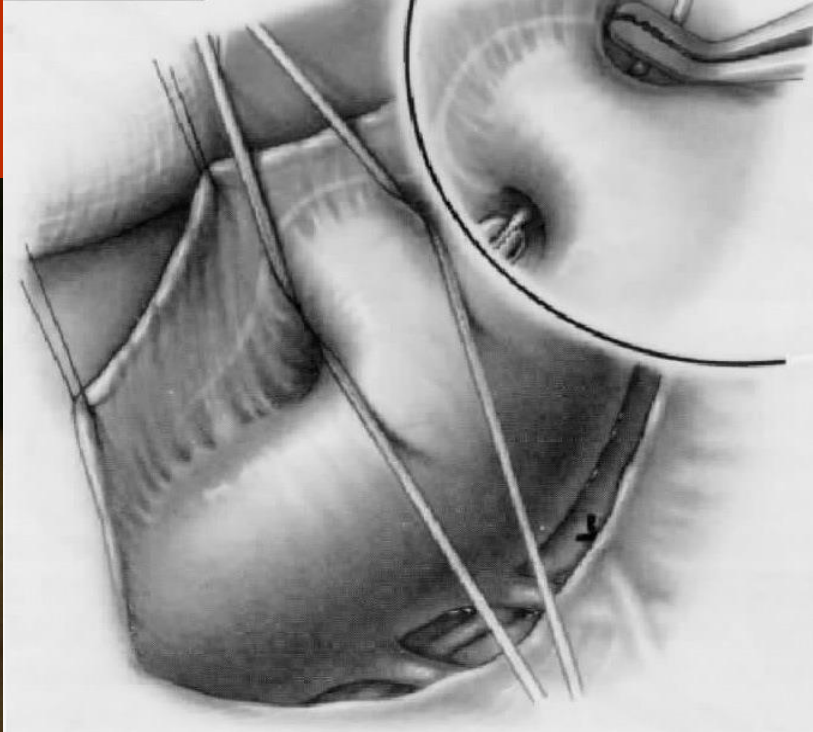


Patent ductus arteriosus

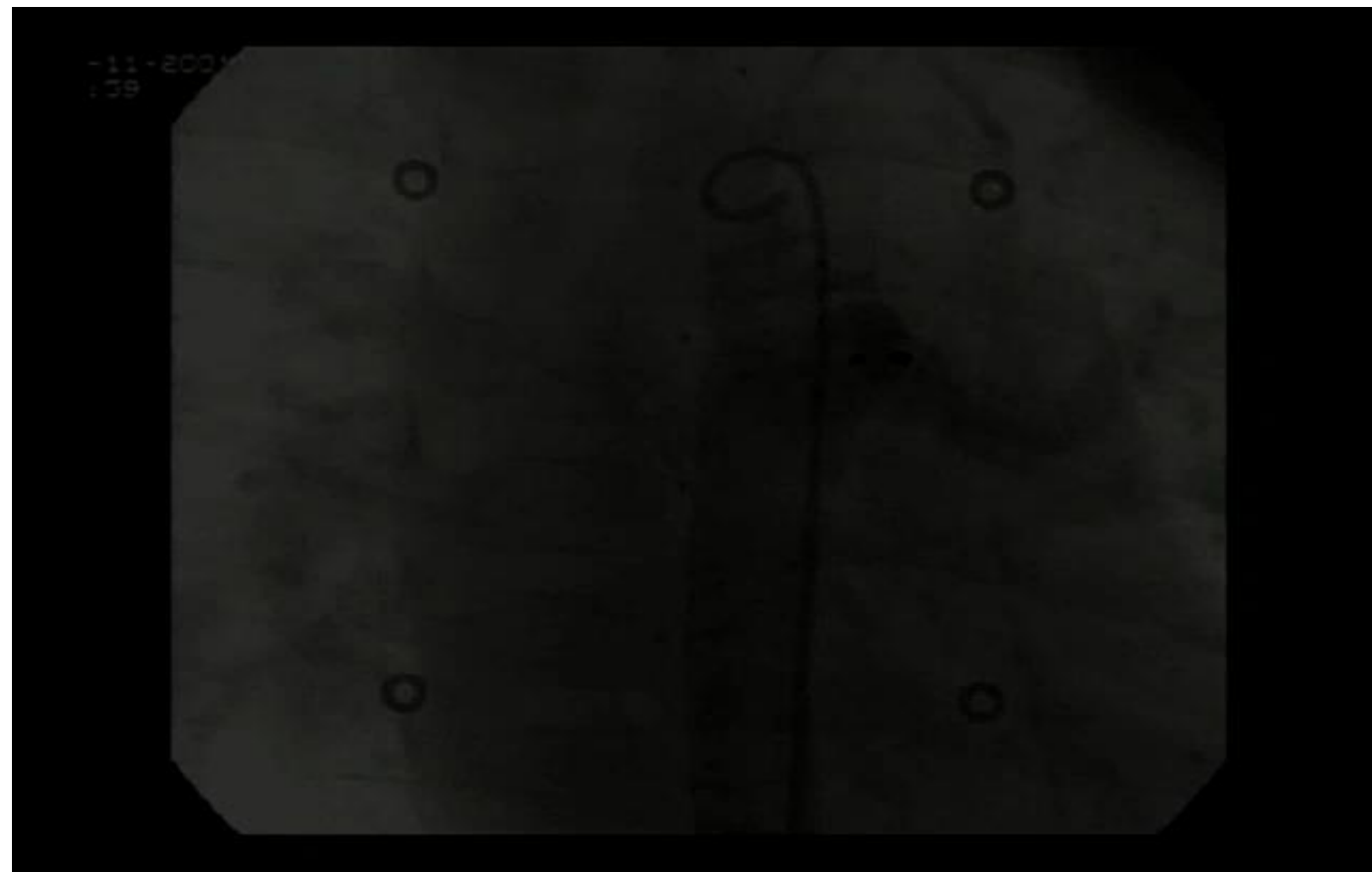
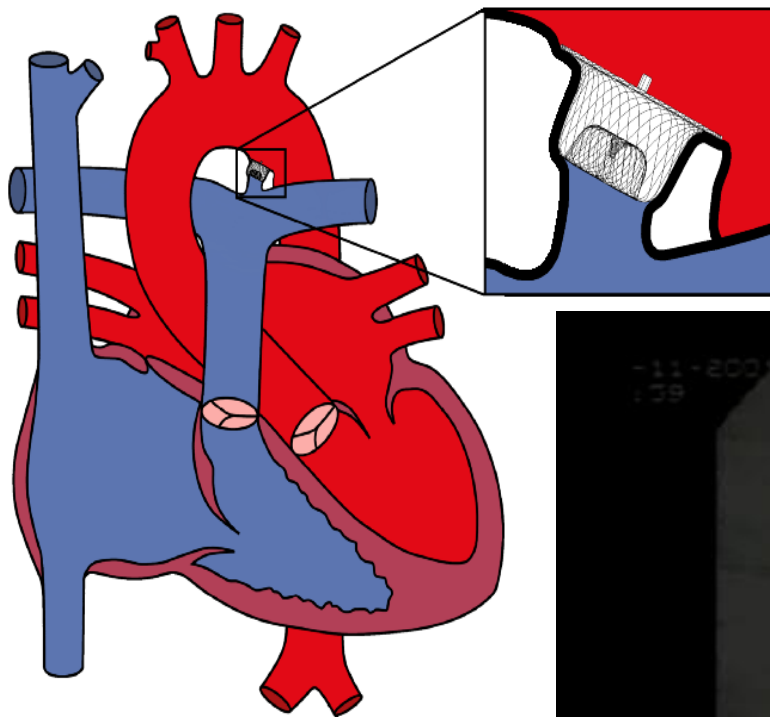


What should we do?

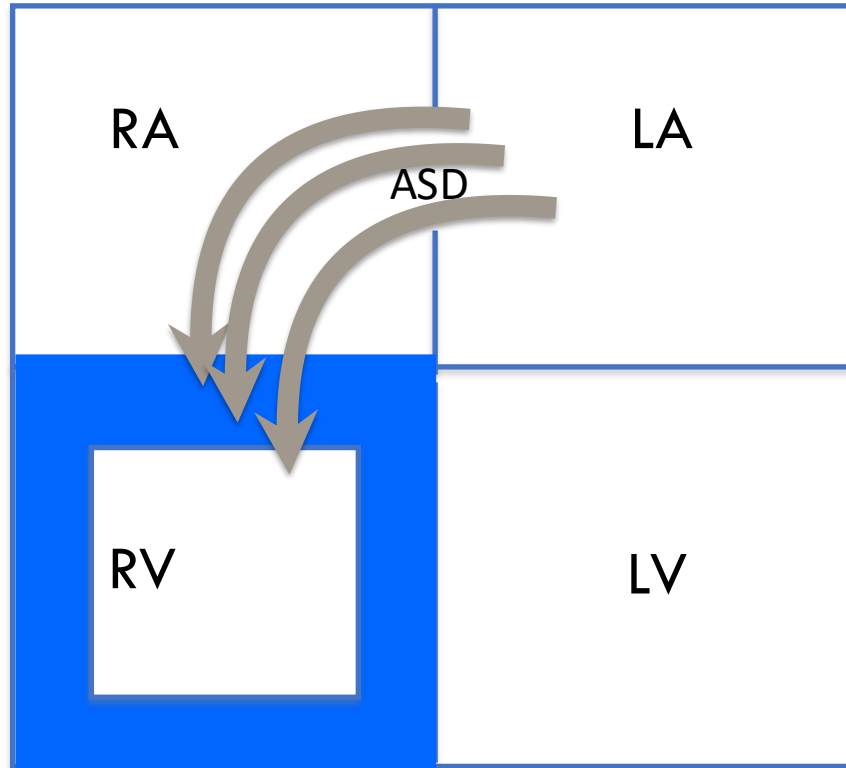
PDA ligation



Intervention with PDA device



Atrial septal defect: ผังหัวใจห้องบนรั่ว



Right heart volume load from ASD

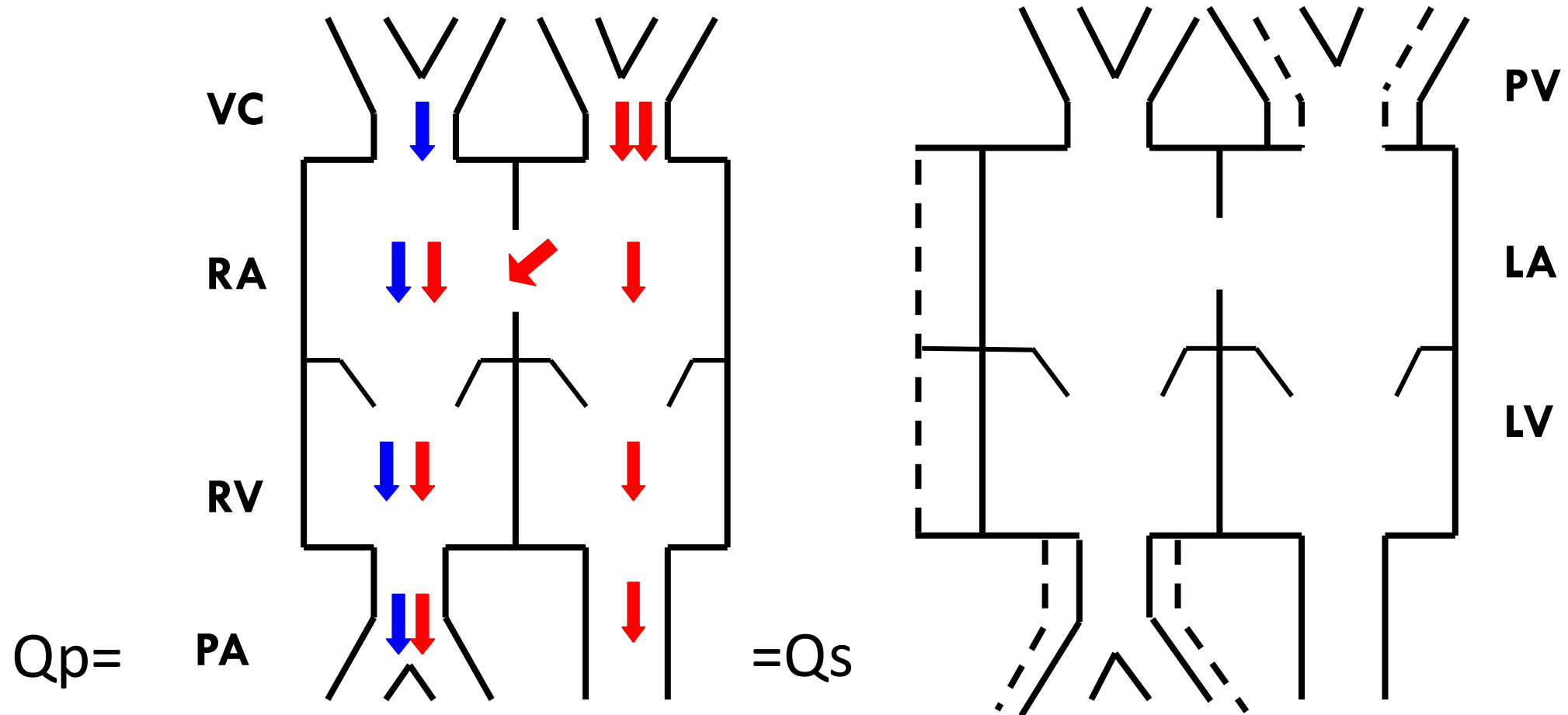
CHF is depend on improved compliance of right ventricle by time

$$\text{Compliance} = \Delta V / \Delta P$$

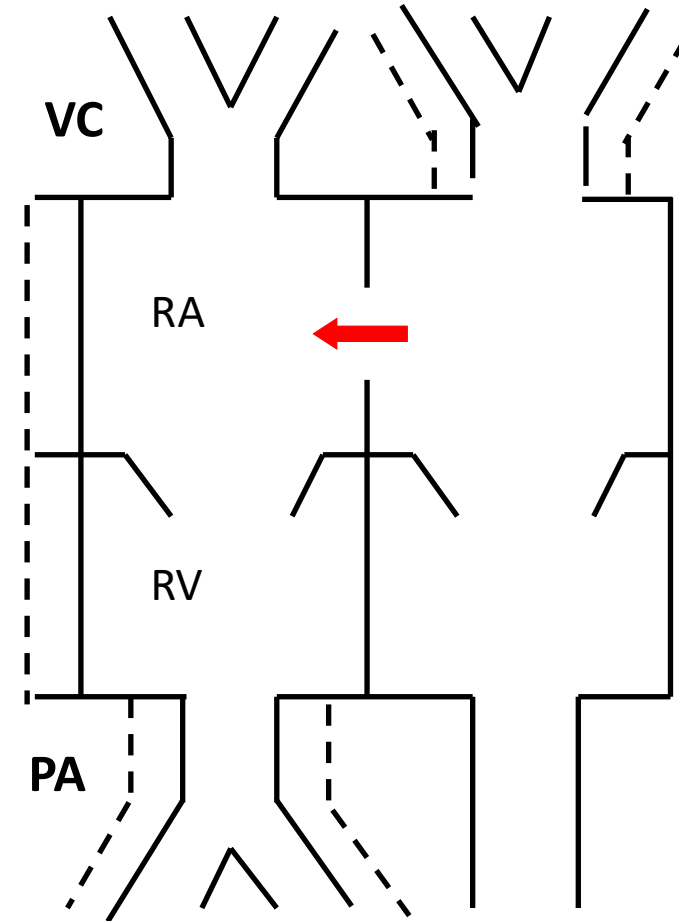
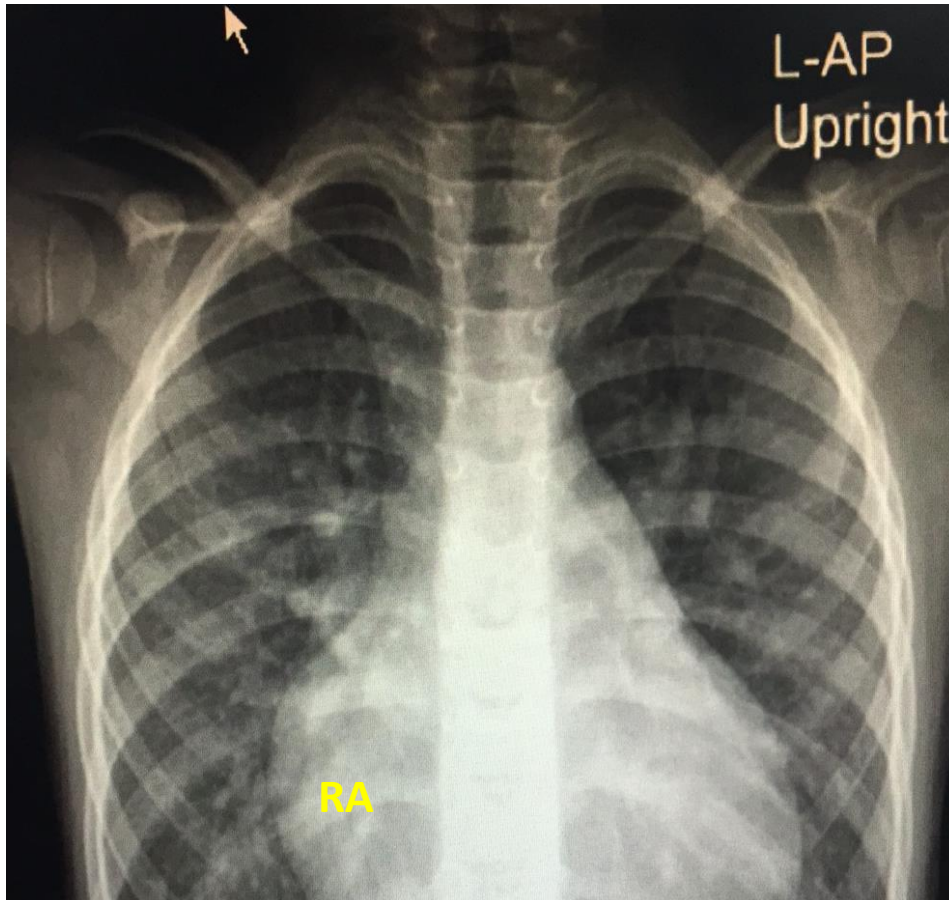
(Greater Ventricular compliance)

▲ V = change in volume
▲ P = change in pressure

Atrial septal defect

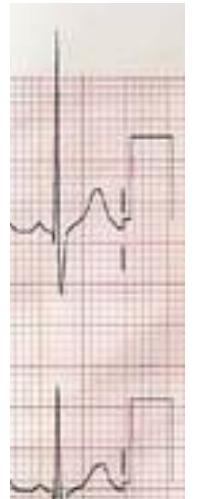
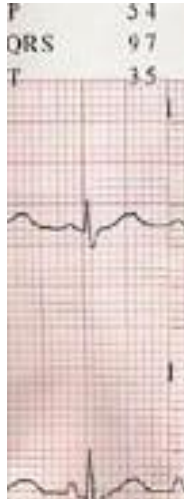


Atrial septal defect



EKG for ASD

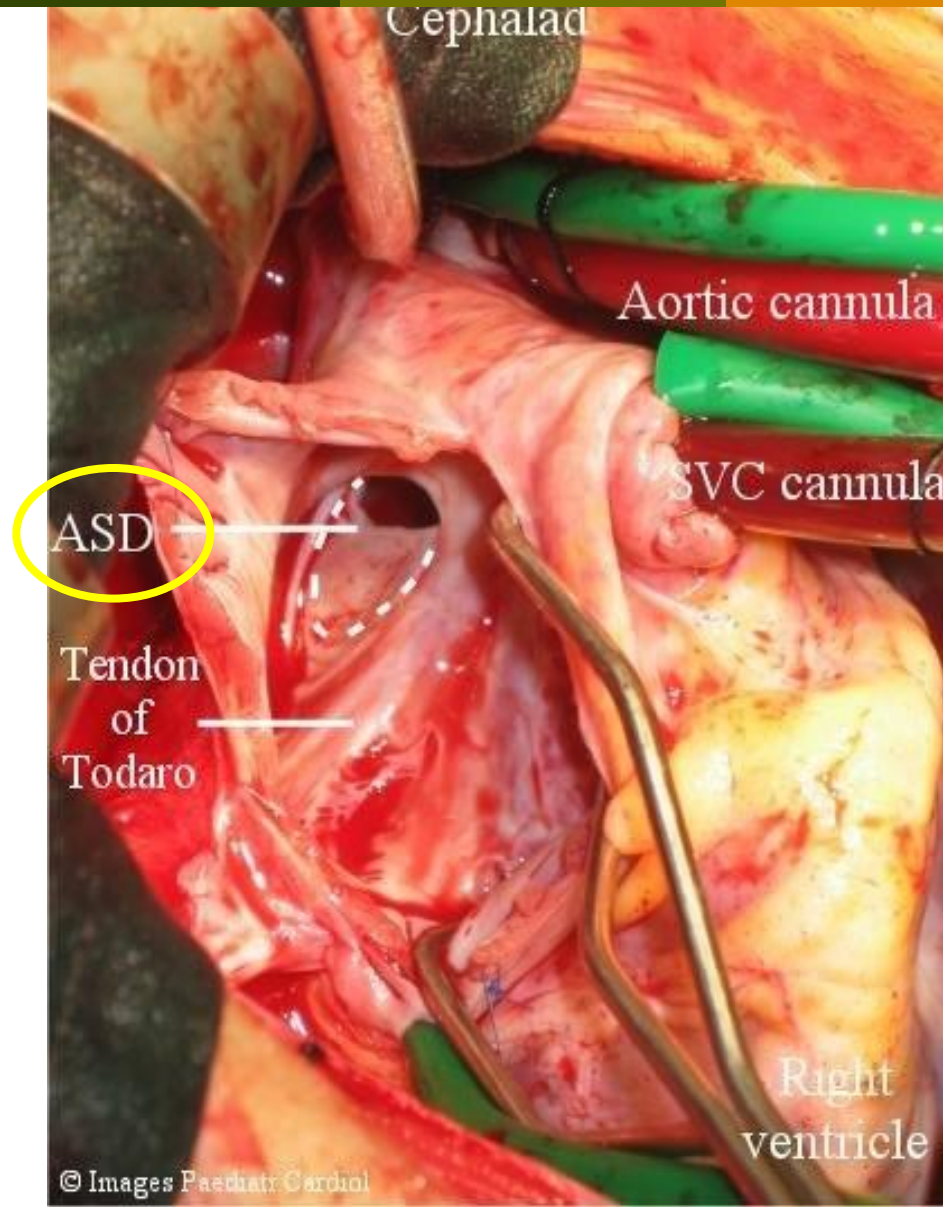
- AV may be moderately prolonged with large defects
- QRS Axis: RAD (90 to 180)
- rsR' or rSR' , incomplete RBBB "Cochetage sign"
- *****Secundum ASD*****



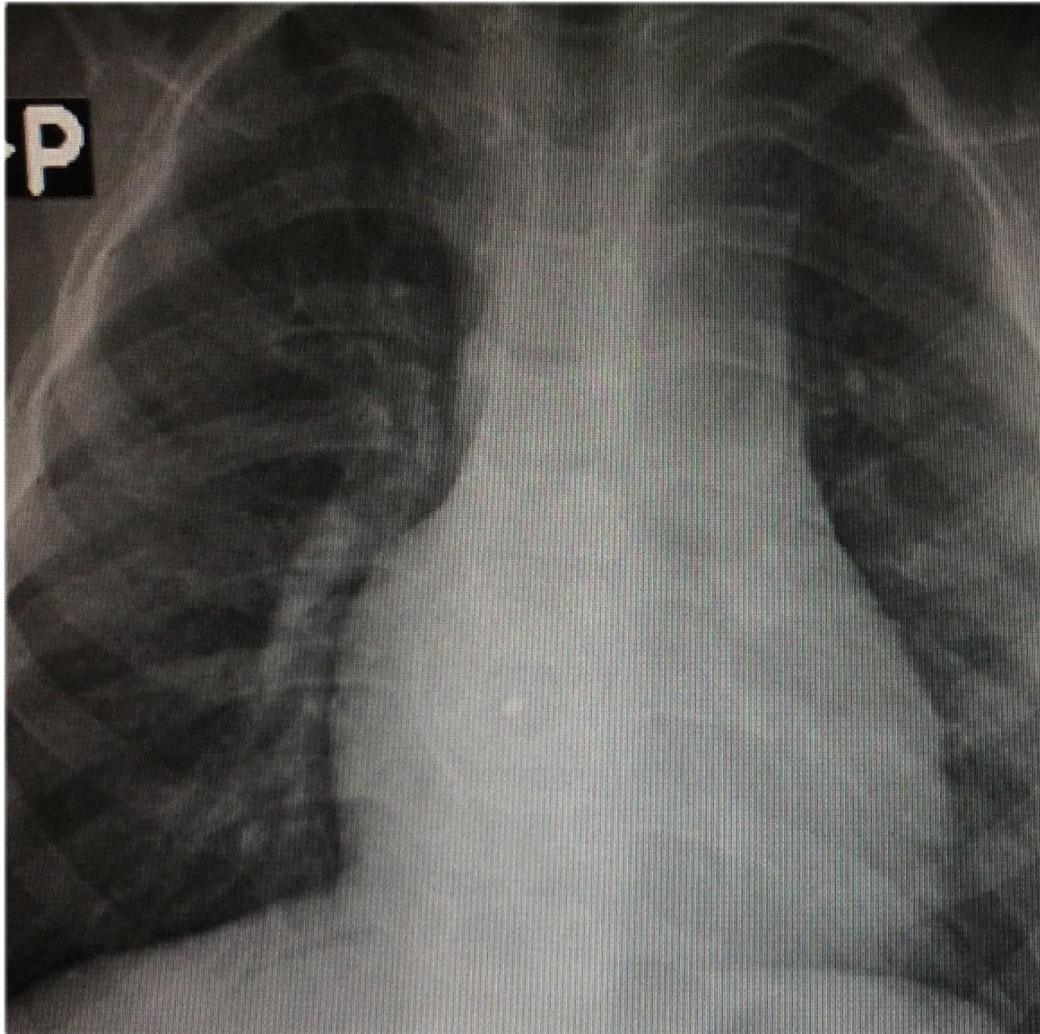
Indication of treatment with Surgery or device

- Symptomatic
- Hemodynamic significant shunt
 - $Q_p:Q_s > 1.5$, right-sized volume load

ASD Surgical Closure



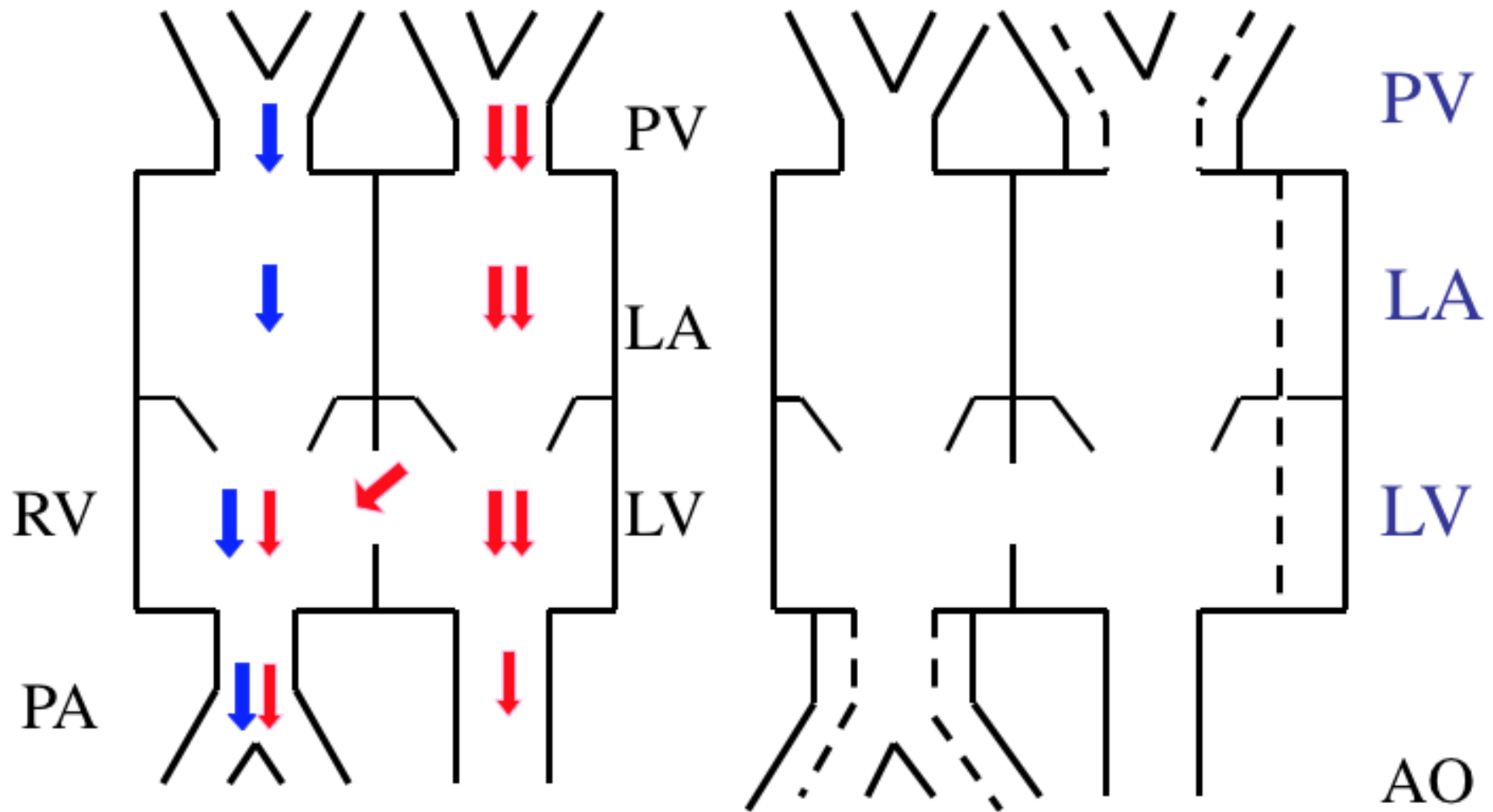
Intervention with ASD device



Take home message

	RAE	RVH	LAE	LVH	PVM	AO
ASD	Y	Y	N	N	Y	N
VSD	N	N	Y	Y	Y	N
PDA	N	N	Y	Y	Y	Y

RAE: Right Atrial Enlargement, RVH: Right Ventricular Hypertrophy, LAE: Left Atrial Enlargement
, LVH: Left Ventricular Hypertrophy, PVM: Increase Pulmonary Blood Flow, Ao: Aorta



โรคใด?

1 patent ductus arteriosus.
 3 Ventricular septal defect

2 Atrial septal defect
 4 Tetralogy of fallot

Thank you for your attention

- พญ.กาญจรัตน์ ว่องไวทวิวงศ์ ภ.กุมารเวชศาสตร์
- หัวข้อ Common congenital heart disease
- <https://forms.gle/d1Vk64EfVWtuDRKaA>

